



GOOD PRACTICE GUIDE:

Digital Validation

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Implementation
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Regulatory Guidance

Challenges and
System Governance



GOOD PRACTICE GUIDE:

Digital Validation

Disclaimer:

This Guide aims to provide practical, risk-based best practices for defining, implementing, and managing Digital Validation Tools (DVTs) within a regulated environment. This Guide is created and solely owned by ISPE. It is not a regulation, standard or regulatory guideline document. ISPE cannot ensure and does not warrant that a system managed in accordance with this Guide will be acceptable to regulatory authorities. Further, this Guide does not replace the need for hiring professional engineers, technicians, or consultants.

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Preface

The adoption of Digital Validation Tools (DVTs) in the life sciences industry highlights an increasing commitment to data integrity, operational efficiency, and regulatory compliance. However, transitioning from traditional, paper-based validation to a digitally integrated framework remains a complex challenge. Organizations often face obstacles with implementation, governance, system integration, and regulatory adherence. The evolving landscape of global regulations, the variety of available technologies, and the heightened auditor expectations further complicate this transition, making structured guidance and industry-wide collaboration essential.

This Good Practice Guide (GPG) offers practical, risk-based best practices for defining, implementing, and managing DVTs within a regulated environment. Developed by a global team of experts in validation, regulatory compliance, quality assurance, and digital transformation, this Guide presents real-world insights, case studies, and structured methodologies to facilitate the successful integration of digital validation into business operations. It explores regulatory considerations, system selection, data management strategies, and compliance frameworks, equipping organizations with the necessary tools to enhance validation efficiency while maintaining regulatory rigor.

Bridging Innovation and Compliance

Successfully implementing a digital validation strategy requires more than just technology—it demands a cultural shift, cross-functional collaboration, and alignment with organizational objectives. This Guide fosters a shared understanding among quality professionals, validation experts, regulatory authorities, and technology providers, ensuring a holistic approach to digital validation. It serves as an essential reference for organizations at all stages of their digital validation journey, whether they are beginning their transition or seeking to optimize existing systems. As regulatory expectations evolve and technology advances, the ISPE Digital Validation Community of Practice remains committed to supporting continued innovation, collaboration, and knowledge-sharing, ensuring this Guide remains an indispensable resource for years to come.

This Guide is aligned with the concepts and framework of the *ISPE GAMP® 5 Guide: A Risk-based Approach to Compliant GxP Computerized Systems (Second Edition)*.

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The collective effort, along with insights from regulatory bodies and technology specialists, have positioned this Guide as a leading reference for validation professionals worldwide.

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Table of Contents

1	Introduction	7
1.1	Evolution of Digital Validation.....	8
1.2	Purpose.....	9
1.3	Scope.....	10
1.4	Structure of the Guide.....	11
2	Current Scope and Advantages of Digital Validation Tools	13
2.1	Case for Quality ⇨ Case for Change.....	15
2.2	Program Governance.....	17
2.3	Effective Change Management.....	17
2.4	Process Ownership for DVT	18
2.5	State of Digital Ecosystem	18
3	Digital Validation Concepts.....	21
3.1	Digital Validation Tool – What It Is/How to Define	21
3.2	Intended Use.....	22
3.3	Data Integrity.....	23
3.4	Culture Change, Organizational Change Management, and Knowledge Management	26
3.5	Data Accessibility	29
3.6	Digital Validation Integration	31
4	Crafting a Compelling Business Case for a Digital Validation Tool.....	39
4.1	Introduction	39
4.2	Highlight the Benefits	41
4.3	Cost/Benefits/Overall Added Value Cost-Benefit Analysis	42
5	Digital Validation Tool Implementation.....	45
5.1	Introduction	45
5.2	Creating Culture for Digitalization	45
5.3	Planning and Developing Implementation Strategies	48
5.4	Stakeholder Engagement	49
5.5	Communication Channels.....	51
5.6	Project Execution	51
5.7	Digital Validation Tool Lifecycle and Fitness for Intended Use.....	52
5.8	Deployment and Integration.....	54
6	Digital Validation Tool Governance.....	57
6.1	Introduction	57
6.2	Governance Team.....	57
6.3	Governance Process	58
6.4	Performance Monitoring.....	67
6.5	Accurate and Usable Validation Records and Copies	69

7	Regulatory Considerations, Inspection Readiness, and Implementation.....	71
7.1	Standards and Guidelines.....	71
7.2	Internal Audits and Regulatory Inspection Techniques and Challenges	72
7.3	Success Stories and Unexpected Challenges of Digital Validation and Inspection Implementation	75
7.4	Future Trends and Developments.....	79
8	Appendix 1 – References	81
9	Appendix 2 – Glossary	83
9.1	Acronyms and Abbreviations	83
9.2	Definitions	85

1 Introduction

Over the past decade, the pharmaceutical and medical device industries have seen a steady increase in the adoption of Digital Validation Tools (DVTs), a software program or system designed to replace traditional paper documentation methods. When used appropriately, DVTs help manufacturers adhere to stringent quality standards while maintaining robust audit trails.

In the pharmaceutical and medical device industries, DVTs provide a comprehensive suite of tools, services, expertise, and applications tailored to manufacturers. These resources are meticulously designed to ensure that documents, software, operational infrastructure, and processes not only comply with regulatory requirements such as the FDA's 21 CFR Part 11 [1] but are also optimized to meet the specific requirements of their intended use. These tools cover an extensive range of applications, from drug product manufacturing and medical device production to medical software applications and laboratory operations, ensuring the integrity and quality of products and processes throughout their lifecycle. By enabling rigorous testing and robust documentation, DVTs help achieve and maintain regulatory compliance, enhance patient safety, and support key areas of the Quality Management System (QMS), including knowledge management and change management.

Through their support of GxP compliance and application of industry best practices such as GAMP, these tools aid in sustaining the high standards required in the production and management of pharmaceutical and medical device products.

The trend in DVT use is driven by an emphasis on data integrity, the emergence of Pharma 4.0™ [29] and Validation 4.0 [30], as well as the need for accessible data on the validation state of systems and processes. The industry is also striving for operational efficiency as part of Pharma 4.0. These advancements aim to streamline periodic reviews and enhance understanding of processes and systems, thereby facilitating the application of Quality Risk Management (QRM) and knowledge management in accordance with ICH Q9 [2] and Q10 [3].

Moreover, the ability to remotely review, execute, and approve validation documentation became crucial during the COVID-19 pandemic, enabling businesses to maintain effective operations and accommodate remote audits by regulatory bodies. This shift towards DVTs is reflected in survey results from the ISPE Commissioning and Qualification Community of Practice (C&Q COP), indicating that approximately 75% of businesses have implemented or plan to implement these systems within the next two years [4]. However, the extent of implementation varies across the industry, with some companies using a hybrid approach of paper and DVTs as they roll out the approach across their validation processes and increase their digital validation maturity. This can be an extended process that may take a multiyear roadmap to complete.

As the adoption of these systems increases, there emerges a necessity for guidance on their utilization within regulated environments and the establishment of best practices. Currently, requirements and guidance are provided indirectly through various sources such as:

- *ISPE GAMP® 5: A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition)* [5]
- FDA CDRH Draft Guidance for Industry: Computer Software Assurance for Production and Quality System Software [6]
- 21 CFR Part 11 [1] and EU GMP Annex 11 [7] and Annex 15 [8]
- Case for Quality (MDIC) initiatives [9]
- FDA Discussion Paper, Artificial Intelligence in Drug Manufacturing [31]

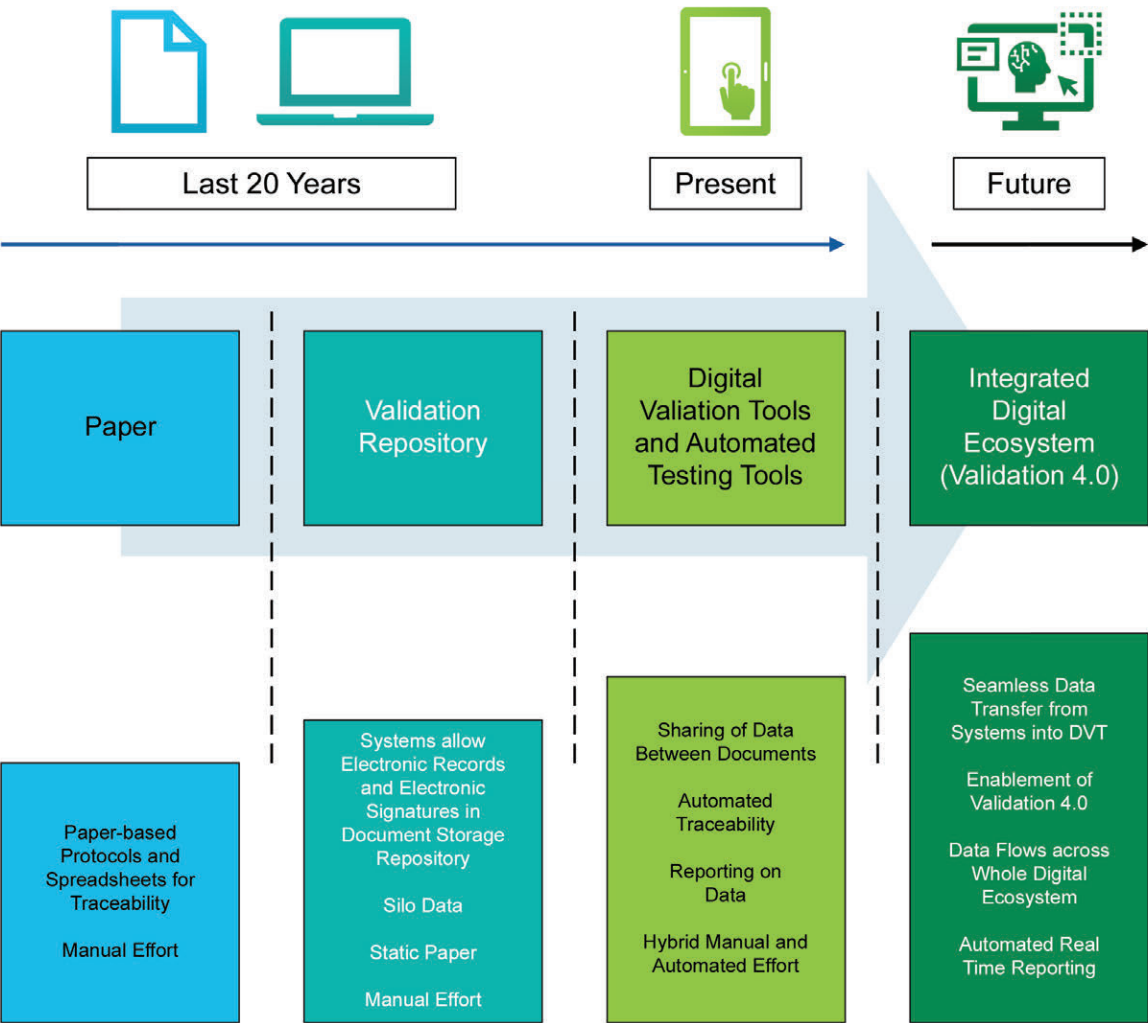
- FDA, Good Machine Learning Practice for Medical Device Development: Guiding Principles [32]
- ISPE GAMP® Good Practice Guide: Enabling Innovation [10]

Because of the lack of direct guidance on the use of DVTs, this Guide aims to present a compliant approach based on current technologies and industry use cases. It also anticipates the future integration of these digital technologies with other systems and their potential applications.

1.1 Evolution of Digital Validation

Since the mid-2000s, the industry has shifted from traditional paper-based validation processes to a digital validation methodology, as illustrated in Figure 1.1. This shift initially involved adopting electronic validation repositories/ Document Management Systems (DMS) that enabled the digital upload, creation, and electronic approval of documentation. The documents were then securely stored within these systems. These platforms provided secure access, storage, and the capability for electronic review and approval, in line with electronic record and signature requirements. However, these systems had limitations, particularly in executing tests directly within them or managing document data, leaving unresolved data integrity and process efficiency challenges associated with paper-based methods.

Figure 1.1: Evolution of Digital Validation



Initially, many of these platforms were adapted DMS or custom-coded solutions that failed to fully address the comprehensive needs of the validation lifecycle. In the last five years; however, there has been a significant shift towards investing in commercially available, customizable off-the-shelf solutions specifically designed to manage validation data digitally, ranging from creation and execution to digital oversight.

These DVTs support digital execution that meets data integrity standards and allow the management and sharing of validation data. This not only enhances efficiency but also ensures compliance in ways that paper-based processes could not. This evolution marks a departure from simply digitizing paper-based processes (“paper-on-glass” approach) to utilizing DVTs for substantial returns on investment and compliance benefits, as discussed in Chapter 4.

In 2023, the ISPE C&Q COP conducted a survey on the adoption of integrated C&Q, focusing on the use of paperless/digital systems for planning, executing, and reporting C&Q activities. According to the survey, shown in Figure 2.1, 74% of respondents anticipate using DVTs for C&Q by 2024, highlighting the robust adoption of these tools within the life sciences sector [4].

As DVT technologies continue to evolve, these tools are paving the way towards achieving the goals of Validation 4.0, where iterative, continuous, and live data integrity by design becomes feasible, offering a value-added approach to managing a product’s lifecycle. Through integration with other systems or utilization of multiple tools, it is now possible to capture and make visible data from various endpoints across the entire validation process lifecycle.

Validation 4.0 represents a transformative approach to validation management in pharmaceutical manufacturing. It moves away from traditional, static upfront and periodic validation methods to a continuous, dynamic, risk management-focused framework. This paradigm emphasizes the demonstration of risk control effectiveness through Quality by Design (QbD) and QRM principles. Complemented by digital technologies, Validation 4.0 enables visibility through data into critical processes, offering real-time insights into the state of control and the state of validation. At higher levels of Pharma 4.0 maturity, Validation 4.0 addresses the limitations of conventional validation approaches, aiming to help organizations transition from static compliance to an agile and proactive quality assurance model that modern products and treatments demand.

The vision of Validation 4.0 centers on establishing a continuous validation framework that supports holistic control strategies and streamlined change management. By organizing and integrating quality considerations throughout the product lifecycle—from development through commercial production, and throughout the entire value chain – from raw material suppliers to downstream patient outcomes—this approach aims to minimize validation activities that do not handle inherent variability and change, and share data across organizational silos to connect a responsive ecosystem. The aspirational goal is to develop a more efficient, interconnected approach to validation management that not only ensures patient safety and product quality but also enables faster, adaptable validation practices to better suit today’s dynamic world.

This Guide discusses how organizations can achieve mature digital validation, and how they can implement these systems in a compliant way (refer to Chapter 7), in addition to highlighting the need to change the culture and mindset of an organization to fully enable the power of these tools, and not “recreate paper-on-glass.”

1.2 Purpose

The purpose of the Guide is to provide specific guidance on how to:

- Create a business case for digital validation
- Successfully implement a DVT, following the GAMP approach
- Provide guidance on how to govern DVTs once implemented
- Provide a regulatory basis for compliant use of these systems

This Guide also discusses key Digital Validation concepts and the importance of creating a digital culture to support implementation of DVTs.

It is designed to offer guidance for the following groups:

- Quality and regulatory groups seeking to understand how DVTs are designed, implemented, and maintained
- Organizations looking to digitalize their validation processes
- Organizations that have implemented DVTs but wish to increase their digital maturity, or scale to processes across the organization
- Organizations seeking to develop their validation processes in alignment with Validation 4.0 principles, while managing product quality in an iterative and continuous manner

Suppliers of DVTs and health authorities who wish to get a better understanding of current industry practices, can also benefit from this guide.

This Guide assumes that the audience has a good understanding of *ISPE GAMP® 5 (Second Edition)* [5], *ISPE GAMP® Good Practice Guide: Enabling Innovation* [10], *ISPE GAMP® Guide: Records and Data Integrity* [11], the three supporting *ISPE GAMP® Data Integrity Good Practice Guides* [12, 13, 14], Draft FDA CDRH CSA Guidance [6], and FDA Guidance for Industry: Process Validation: General Principles and Practices [16].

1.3 Scope

DVTs encompass a range of software applications (one or more integrated) designed for managing and overseeing digital assets essential for the qualification, verification, validation, and assurance of compliance in various life sciences domains.

The scope of this Guide encompasses tools that maintain GxP records, including formal validation plans and reports for process validation, analytical method validation, and other validation activities. It also includes data directly supporting GxP activities, such as process validation data, analytical and cleaning data, and process performance qualification batch data for commercial products. These DVTs require validation.

GxP Computerized system lifecycle records and supporting IT process records do not directly support the medicinal product lifecycle are not required by predicate rules and are not considered GxP records, as defined and described in *GAMP® 5 (Second Edition)* [5], Appendix D9 Software Tools. These GxP system lifecycle and IT process records are out of scope and should be managed according to the guidance in that appendix. If a DVT is used solely to manage such system lifecycle records and IT process records, it is regarded as a GAMP Software Category 1 tool, and validation is not required for that intended use.

DVTs may be used to support computerized system validation activities and can offer efficiency gains in doing so. However, the validation paradigm, along with the GAMP framework and principles, remains unchanged, and the objective of ensuring fitness for intended use can be achieved without the use of such tools.

The scope of this guidance covers the lifecycle of a DVT through concept, project, operation, to retirement.

The use cases cover the whole spectrum of validation, qualification, and verification, including but not limited to, the following:

- Process Validation (including cleaning and sterilization validation/Process Performance Qualification (PPQ))
- Computerized System Validation and Computer Software Assurance
- Facilities, Utilities, Equipment, and systems commissioning and qualification
- Analytical Equipment/Instrumentation Qualification
- Validation of test Methods according to regulations
- Packaging Validation
- Transport/Shipping Validation
- Periodic Review, Requalification, and Validation Planning
- Change Management

There are other potential uses of DVTs, such as engineering documentation repositories; however, these are not within the scope of this Guide.

1.4 Structure of the Guide

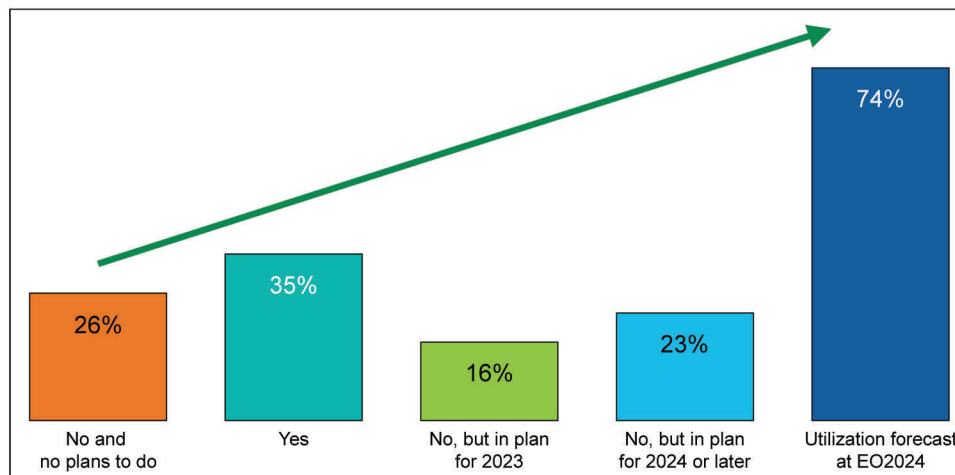
The Guide is structured to cover the lifecycle of a DVT and contains the following chapters:

- Chapter 2: Current Scope and Advantages of Digital Validation Tools – this chapter that discusses the background and current state of DVT use and current and projected advantages.
- Chapter 3: Digital validation concepts – This chapter discusses key concepts that are the foundation of the Guide.
- Chapter 4: Crafting a Compelling Business Case for a Digital Validation Tool – This chapter provides guidance on creating a business case and providing Return on Investment (ROI) for a DVT.
- Chapter 5: Digital Validation Tool Implementation – This chapter provides guidance on implementation and validation of DVTs.
- Chapter 6: Digital Validation Tool Governance – This chapter provides guidance and case studies on how to successfully govern a DVT.
- Chapter 7: Regulatory Considerations/Impact – This provides guidance on compliant use of a DVT.

2 Current Scope and Advantages of Digital Validation Tools

The 2023 survey conducted by the C&Q Community of Practice on the application of Digital Validation Tools (DVTs) in Commissioning and Qualification (C&Q) efforts revealed that approximately one-third of respondents have integrated DVTs into their C&Q workflows. This adoption rate is expected to increase significantly in the coming years, as illustrated in Figure 2.1, which details the industry adoption rate of DVTs.

Figure 2.1: Industry Adoption Rates of DVTs [4] (created by C&Q CoP from 2023 survey data)



The integration of DVTs in C&Q operations is often prioritized due to the substantial ROI that can be achieved, especially in large-scale projects. This topic is explored in Chapter 4. The advantages of employing DVTs over traditional paper-based methods during various validation stages include:

- **Real-time Visibility of Progress:** DVTs offer immediate updates on documentation status, thereby eliminating the need for traditional manual tracking. This enables clear visibility into key metrics, such as the review and approval times of documentation, facilitating the resolution of any approval bottlenecks.
- **Improved or Automated Requirements Traceability:** Modern Risk-based C&Q methodologies require that Process User Requirements (PURs) and Critical Design Elements (CDEs) be traced to their verification points. Manual traceability management is labor-intensive and prone to errors. DVTs minimize this by linking requirements/CDEs directly to verification tests, streamlining document creation, review, and report generation, and enabling faster system handovers. Digital traceability can improve change management by enabling more robust impact assessments and supporting ongoing lifecycle management of traceability as controls evolve.
- **Facilitation of Remote Work:**
 - DVTs enable the remote creation, review, approval, and execution of C&Q documentation. This capability is particularly beneficial for vendor testing during Factory Acceptance Testing (FAT), eliminating the need for a physical presence at vendor sites, a significant advantage during travel-restricted periods such as the COVID-19 pandemic. Augmented Reality (AR)/Virtual Reality (VR) technologies further enhance this by providing visual oversight of test execution and saving video evidence.

- Compared to paper-based reviews, DVTs support more collaborative approaches, enabling real-time document reviews and amendments by teams using collaborative tools like Microsoft Teams, Word, or Excel, complete with compliant electronic signatures. The real-time visibility and availability of validation data can also support ongoing process and product management (such as in deviation investigations, change impact assessments); refer to Chapter 2 for more discussion.
- **Elimination of Paper-related Costs and Compliance Issues:** Managing the voluminous paperwork generated by large projects involves significant storage and control efforts. DVTs reduce these costs by digitalizing evidence storage, adhering to ALCOA++ principles, ensuring audit readiness, and facilitating the controlled destruction of paper copies when “true copy” principles are applied. Additionally, DVTs mitigate the risks of loss and falsification associated with paper documents through robust audit trails and access controls.
- **Cost savings Associated with Energy and Environmental Footprint:** DVTs enable cost savings by eliminating the need to maintain warehouses for storing paper records, which require electricity, temperature, and humidity-controlled systems along with other services for compliance and accessibility. With digital records, these capital projects and storage spaces become obsolete, providing more space to expand manufacturing capacity rather than paper storage. DVTs also contribute to sustainability by reducing the need for paper and ink, which leads to large CO₂ reductions.
- **Harmonization of Validation Processes:** By implementing standardized templates, processes, and workflows, DVTs help unify validation approaches across an organization, ensuring adherence to Standard Operating Procedures (SOPs) and compliance throughout the validation lifecycle.

These advancements reflect the shift towards more compliant, efficient, and cost-effective validation processes through the adoption of DVTs.

The implementation of DVTs in large projects significantly enhances scheduling, cost-efficiency, and compliance. Life science companies, faced with the dual challenges of reducing costs and accelerating the market delivery of potentially life-saving medications, can leverage DVTs to meet these goals more effectively.

DVTs are used across various validation processes within life science organizations, improving efficiency and compliance. Key areas where DVTs have been effectively applied include:

1. **Cleaning Validation:**

- DVTs streamline the creation of cleaning validation protocols and reports, offering quick access for operators via tablets on-site. These tools specify the exact locations for swabbing from a worst-case scenario perspective, significantly reducing downtime.
- Modern DVTs feature Application Programming Interface (API) capabilities that facilitate integration with Laboratory Information Management Systems (LIMS), automating the population of cleaning validation and verification reports with approved sample results.

2. **Process Validation (PV):**

- DVTs allow for the digital execution of PV protocols in the field, streamlining the verification of key sample collections, maintaining the custody chain of samples to LIMS, and automating the reporting of results, which accelerates the generation of PV reports.
- Trends in Critical Process Parameter (CPP) results can be integrated into PV reports from data lakes (e.g., data historian systems) to ensure compliance with Normal Operating Ranges (NORs).

3. **Commissioning and Qualification of Facility, Utilities, Equipment, and Systems:**

- DVTs support automated software testing through tools or APIs that interface with automation tools, improving both efficiency and accuracy.

4. **Continuous Process Verification (CPV):** DVTs are instrumental in streamlining CPV efforts, offering robust tools for continuous monitoring and verification, ensuring ongoing compliance and process optimization.

These applications of DVTs not only support life science companies improve operational efficiencies but also play a crucial role in ensuring that life-saving drugs are brought to the market faster and in compliance with regulatory standards.

This Guide discusses the approaches to using DVTs in compliance with applicable regulatory requirements in Chapter 7.

2.1 **Case for Quality ⇌ Case for Change**

The historical approach to validation processes in life sciences has evolved gradually but significantly, shifting from manual, paper-based processes to digital and automated platforms, marked by regulatory milestones such as the FDA 21 CFR Part 11 in 1997 [1]. This evolution, driven by the need for increased efficiency, accuracy, and compliance in a highly regulated environment, has accelerated in recent years. However, the adoption of advanced technologies and methodologies has been slower than expected due to a cautious approach towards regulatory interpretations and quality cultures resistant to change.

To meet the current and future demands of the industry, a fundamental shift in the quality mindset is needed. Organizations must embrace a more dynamic, adaptive approach to quality management, incorporating technological advancements to enhance compliance, improve product quality, and drive innovation in qualification, verification, and validation processes. This requires a cultural transformation that prioritizes continual improvement, risk management, and digitalization as core components of quality assurance that will further evolve the digital validation transformation within an organization.

Chapter 5 outlines methods for implementing DVTs, addressing the transformative shift from traditional, manual processes to a fully integrated digital platform. The goal is the establishment of an organizational culture that fosters digital innovation and adaptability, supported by a digital mindset and an environment conducive to technological transformation. The chapter details the planning and development of implementation strategies, stakeholder engagement, and project execution while considering various deployment approaches and integration with existing systems.

Validation of the DVT platform is a key step to ensure that the system's functionality aligns with organizational goals and meets quality and compliance standards. Furthermore, the transition involves careful planning for decommissioning legacy systems and transferring content to the new digital validation platform. Chapter 5 highlights challenges and lessons learned, reinforcing the notion that strategic planning and organizational readiness are paramount for a successful digital transformation. This comprehensive approach not only aims to streamline validation processes but also foster a culture of continual improvement and innovation within the organization.

Chapter 5 introduces a digital validation maturity model, outlining the transformation from Level 1 to Level 5. This progression requires significant cultural shifts within an organization, with three key quality culture transformations that need to be managed throughout this journey.

2.1.1 From Reliance on Traditional Methods to Embracing Digital Innovation

At Level 1, organizations are deeply rooted in traditional, paper-based methods and waterfall project management approaches. The shift towards digital innovation requires a fundamental change in mindset from all levels of the organization. This transformation involves moving beyond established practices to embrace digital tools and methodologies. This cultural shift is critical for adopting Agile practices, digital documentation, and automated processes that characterize higher levels of digital maturity.

A practical step an organization can take to facilitate the shift from reliance on traditional methods to embracing digital innovation is to implement pilot programs focused on digital tools and Agile methodologies. Starting with a small, manageable project allows the organization to experiment with digital documentation, Agile practices, and automation in a controlled environment. This approach demonstrates the tangible benefits of digital innovation while building confidence and acceptance across different levels of the organization. Supporting these programs with training sessions and workshops alongside these pilot programs can enhance understanding and acceptance, providing employees with the skills and knowledge needed to navigate the new digital tools and methodologies. This step serves as a foundational move towards changing the organizational mindset, fostering a culture that values and seeks digital innovation.

2.1.2 From Risk Aversion to Risk Management

Many organizations follow procedures that emphasize compliance overactive risk management. As they progress, a cultural shift towards a proactive approach to risk management is necessary. This involves integrating risk-based decision-making processes, where actions are prioritized according to real-time risk assessments. Cultivating a culture that understands, accepts, and actively manages risk is essential for leveraging the full benefits of digital capabilities in validation processes.

To foster a shift from risk aversion to effective risk management, organizations can implement a comprehensive risk management training program. This program should focus on educating employees about the principles of risk management, including how to identify, assess, and prioritize risks based on their potential impact. Using real-life scenarios and simulations can provide hands-on experience, helping employees apply risk management strategies and enhance their ability to make informed decisions in real-time situations. Additionally, establishing a cross-functional risk management team can facilitate the sharing of best practices and promote a more integrated approach to risk management. This team can serve as risk management champions, encouraging a culture where risk is strategically managed to drive innovation and operational excellence.

2.1.3 From Siloed Operations to Integrated Collaboration

In the early stages of digital maturity, organizations often experience siloed operations, with limited collaboration between departments. This can lead to inefficiencies, such as disconnected data flows or missed integration opportunities, preventing the creation of a cohesive digital ecosystem. As organizations evolve, a significant shift occurs towards integrated collaboration where cross-functional teams work together seamlessly driven by shared digital platforms and Agile methodologies. This change is both organizational and cultural, requiring the breakdown of traditional barriers to communication and collaboration.

Emphasizing transparency, shared objectives, and open communication is vital for fostering a culture of collaboration that supports the agility and rapid adaptation required at higher maturity levels. To bridge the gap from siloed operations to integrated collaboration, organizations can initiate cross-functional projects that require cooperation between departments. These projects can be specifically designed to address complex problems that no single department can solve independently, thereby requiring input, expertise, and cooperation from multiple teams. Accompanying these projects with the implementation of shared digital platforms can further enhance communication, project management, and documentation in real-time. By promoting the use of Agile methodologies within these cross-functional teams, the organization encourages flexibility, rapid iteration, and a focus on shared goals and objectives. Regularly scheduled cross-departmental meetings to share progress, challenges, and successes can improve transparency and mutual understanding, fostering a collaborative culture essential for digital transformation.

Managing these cultural shifts involves strategic planning, ongoing communication, education, and strong leadership commitment. It requires a consistent effort to engage employees, aligning them with the new values and practices that underpin a digitally mature organization. This journey, while challenging, is essential for organizations aiming to achieve higher levels of efficiency, compliance, and innovation in validation processes.

2.2 Program Governance

Governance plays a critical role in the success of any project, and this is no different when it comes to adopting DVTs. By strategically leveraging different governance approaches, organizations can effectively address the challenges associated with DVT adoption (refer to Chapter 6).

Top-Down Governance:

- Set clear direction
- Allocate the Correct Resources
- Align with Strategic Objectives
- Policy and Procedure Development and Change

Bottom-Up Governance:

- Employee Involvement and Engagement
- Innovation and Creativity
- Feedback and continual improvement
- Reduce Resistance to Change

By combining these key principles an organization can create a more comprehensive and inclusive change management strategy when introducing DVTs. The top-down approach addresses the need for strategic direction, proper resource allocation, and the development of policies, while the bottom-up approach fosters employee engagement, innovation, and adaptability. Together, they make the change process smoother and more successful.

2.3 Effective Change Management

For organizations that have successfully implemented DVTs and use them daily, maintaining an effective and efficient change management process is important. Once deployed, the DVT will undergo many changes and to maintain the validated state of the DVT there are several factors to consider:

- **The Validated State of the DVT:** This includes version upgrades, new functionality patches to existing versions, Commercial Off-The-Shelf (COTS) solutions, configurable or custom interfaces
- **The Validated State of the Processes:** The processes being used within the DVT must also be maintained

Maintaining the validated state of the DVT itself is relatively straightforward as any changes require proper change control and verification to ensure that new functionality does not impact existing validated features. However, the business processes within the system can be more complex.

To maintain the validated state of business processes within the DVT, a Risk-Based Change Management approach can be employed. This ensures changes are implemented effectively and efficiently. This process includes the following steps:

- Identify potential changes that may be introduced to the validated business process, e.g., introduction of new documents, templates, business processes or documentation packs
- Assess the risk associated with these changes, e.g., introduction of a new template may impact other areas of the business process and therefore measures should be taken to control this impact
- Change Planning and Implementation, e.g., develop change plans for the different levels of risk, with a defined list of actions and responsibilities

2.4 Process Ownership for DVT

The Process Owner plays a crucial role in ensuring that the business process runs smoothly and efficiently and this is no different when it comes to the use of a DVT. Although the tool itself is classified as an IT system and validated as such, the processes configured within the tool are typically owned by the engineering and validation teams. Therefore, it is important that the process owner is a member of one of these teams to ensure the following:

1. **In-depth Understanding of the DVT:** The process owner must have a deep understanding of both the tool and the process it supports. Without this knowledge, the process owner may struggle to make informed decisions or effectively address emerging issues, leading to inefficiencies, errors, and ultimately, a breakdown in the process.
2. **Effective Decision-Making:** The process owner is often required to make critical decisions related to the process, such as prioritizing improvements, allocating resources, or resolving bottlenecks. If the person in this role lacks the necessary expertise, poor decisions may be made, hindering the process's overall performance.
3. **Stakeholder Buy-In:** Implementing changes or improvements to a business process often requires buy-in from various stakeholders across the organization. The process owner should have a level of influence that enables good partnership with key stakeholders and addresses any resistance to change. This will help prevent delays or the rejection of necessary improvements.
4. **Continual Process Improvement:** A capable process owner should continually seek opportunities to innovate and optimize processes to improve efficiency and effectiveness. Without a forward-thinking mindset, the BPO may overlook valuable opportunities for process improvement.
5. **Minimizing Process Failure Risk:** Assigning the right person to the process owner role significantly decreases the risk of process failure due to lack of understanding, commitment, or skills mismatch. The right process owner can ensure that processes set up within the DVT are aligned with business goals and are optimal for business use.

2.5 State of Digital Ecosystem

Many current DVTs deployed in organizations or commercially available are still in a transitional phase. They represent an advancement from traditional paper-based and “paper-on-glass” methods to digital, yet not fully embrace a total digital transformation. DVTs often replicate paper-based processes in digital form, including documents such as Installation Qualification (IQ), Operational Qualification (OQ), and Requirements Specification/ User Requirement Specifications (RS/URS)¹, while also attempting to adopt more modern approaches, such as those outlined in *ISPE GAMP® 5 (Second Edition)* [5].

¹ Both terms RS and URS are widely used, and this document uses “RS” for consistency with *ISPE GAMP® 5*.

This Guide aims to remain flexible and does not intend to prescribe any single set of terms to the exclusion of others. It does not use traditional pharmaceutical process validation terminology such as IQ/OQ/PQ to describe system lifecycle activities.

These digital systems have inherited structures from their paper-based predecessors, largely due to digital validation providers adapting existing paper processes upon client request. While DVTs incorporate some automation and workflows, the data within these systems remains largely static, necessitating extraction for analysis or trending. The integration capabilities of these systems are often limited, with few APIs for data sharing, and lack of unified access protocols across platforms like DVTs, DMS, Engineering Information Management Systems (EIMS), and asset management systems.

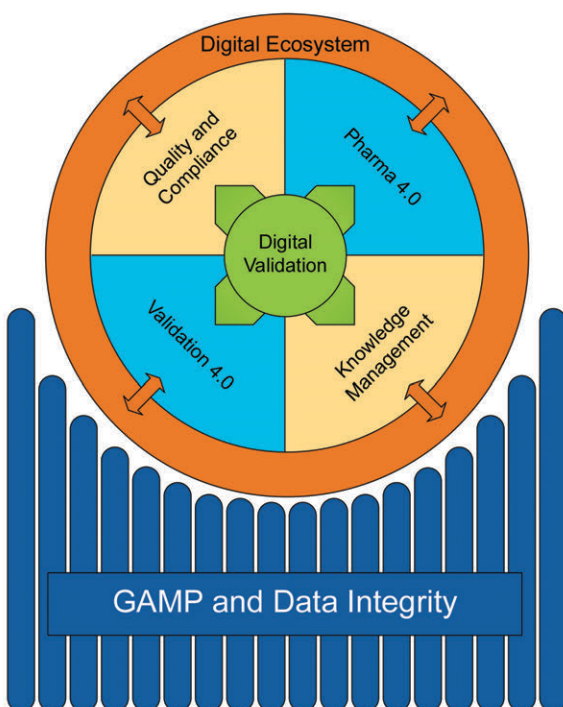
Despite being designed for digital operations, the change management processes in some DVTs still require manual intervention for updates to individual components such as URS or OQ documents. Moreover, certain systems are focused narrowly on specific areas such as computer system validation, neglecting the broader spectrum of validation needs.

Companies often deploy these DVTs in isolation for two main reasons: the inherent stand-alone design of many leading DVTs and a siloed organizational mindset. This mentality, driven by departmental ownerships and a lack of centralized governance over all digital systems, impedes the creation of a cohesive digital ecosystem. Additionally, the presence of multiple software versions and providers within a company, often a byproduct of mergers and acquisitions rather than inadequate governance, further complicates efforts to create an interconnected digital landscape.

To counteract these challenges, comprehensive data mapping should precede any implementation. This process should consider all existing systems, their versions, data access capabilities, and data flows between them. This preparatory step is essential for developing an integrated digital ecosystem and dismantling operational silos.

Currently, a true digital validation ecosystem is more an aspiration than a reality. An example of a model of a digital ecosystem is shown in Figure 2.2, where quality and compliance, knowledge management, Pharma 4.0, and Validation 4.0, are enabled by a DVT, supported by GAMP and data integrity principles.

Figure 2.2: Digital Ecosystem



The reason that this digital ecosystem has not yet become a reality is largely due to these integration and standardization challenges. However, a key objective of this Guide is to promote innovation and integration towards realizing a cohesive digital validation ecosystem. Further details on implementing DVTs are found in Chapter 5.

3 Digital Validation Concepts

3.1 Digital Validation Tool – What It Is/How to Define

Digital validation refers to the use of DVT to remove and/or replace paper-based steps in validation processes. The DVT is a software program or system designed to replace traditional paper-based documentation methods, such as managing requirements, protocols, test execution steps, validation evidence, test data, and test results. The digital solution supports establishing and enforcing organization-wide standards, streamlines the validation process by automating data capture, analysis, and report generation, which in turn improves efficiency, reduces errors, and ensures regulatory compliance. As a result, DVTs help manufacturers adhere to stringent quality standards while maintaining robust audit trails.

The shift towards virtual or remote audits and inspections, alongside growing FDA support for digital tools, demonstrates that DVTs are not only feasible but superior to paper-based methods. A DVT can be applied to any validation activity currently performed on paper. The transition to digital validation offers significant advantages on validation activities such as:

- **Data Entry:** Paper-based workflows often require data entry into word processors or spreadsheets, followed by printing and archiving. These time-consuming tasks divert skilled QA staff from high-value activities. Digitizing the data collection process eliminates these steps, enabling practitioners to focus on tasks such as risk assessment. DVTs also provide real-time visibility into the progress of the validation effort and can easily generate reports and metrics for faster decision-making.
- **Routing:** Manual routing of validation documents via email or inter-office mail can cause delays and make it difficult for reviewers and developers to access the latest versions. These delays hinder the development process. In contrast, electronic workflows simplify document sharing, ensuring that teams can enforce requirements consistently throughout the development lifecycle and avoid last-minute issues. Electronic routing also allows for tracking document ownership and reducing the risk of lost files.
- **Analyzing:** The process of reviewing and searching for a specific abnormality in validation documents is time consuming and prone to human error. Machine learning algorithms can quickly search, sort, and analyze digital data, identifying trends within minutes. This allows reviewers to focus their expertise on addressing anomalies and errors effectively.
- **Reviewing:** Paper documents allow only one reviewer at a time, requiring sequential reviews for effective version control. This process can be complicated by the presence of multiple document copies. Digital validation enables simultaneous authorship and editing by multiple reviewers, with version controls that track who made which changes and when, while automatically creating an audit trail.
- **Notification:** The paper-based review process is slow, requiring reviewers to wait for physical documents. Digital validation tools offer real-time alerts for document readiness and task assignments, allowing reviewers to accept or decline tasks immediately. This enables efficient task reassignment and minimizes delays.
- **Trace Matrix:** In paper-based workflows, traceability matrices are often neglected until the end of the validation process, risking errors due to complexity and time constraints. Digital workflows integrate traceability throughout the project, enhancing validation and software testing efficiency and accuracy. Digital trace matrices can be maintained more easily and can be used for future change management, rather than simply as documentation for completed validation.
- **Deviation:** Handling deviations and exceptions is time consuming and difficult to track in paper-based systems. Digital workflows streamline deviation management, allowing for concurrent handling and faster execution. Digitization ensures traceability of each deviation and validation change, improving issue resolution and maintaining product quality compliance.

- **Source of Truth:** In paper-based workflows, physical copies are required, increasing the risk of loss or security issues. Digital validation tools provide a secure, centralized “source of truth” with role-based access controls. Virtual access facilitates global collaboration, saving time and enhancing security.
- **Periodic Review:** Manual workflows complicate the timely scheduling of reviews and processes for maintaining the validated state. Paper documents often lack a unified calendar for tracking deadlines. Digital validation workflows leverage dynamic calendars and real-time alerts to enable on time reviews and approvals. Also integrating AI and machine learning can improve content consistency within the DVT by enabling functionality such as similarity analysis to identify inconsistencies and consolidate requirements

Defining a DVT involves specifying the system’s capabilities and its compliance with industry standards. Key features may include secure electronic signatures, customizable workflows, real-time data analysis, and comprehensive reporting functionalities. The tool must also comply with relevant regulations such as FDA 21 CFR Part 11 [1], and EU GMP Annex 11 [7]. By setting these parameters, organizations can ensure that the tool not only improves efficiency but also meets all legal and quality requirements.

This digital approach accelerates the validation process while significantly reducing resource consumption and costs compared to paper-based methods. A primary goal of DVTs is to transition from document-centric to data-centric validation, integrating all relevant data processing functions, including calculations and statistical analysis, into the tool.

3.2 Intended Use

DVTs are designed for use across all areas of validation of facilities, utilities, equipment, systems, and processes within the pharmaceutical, biopharmaceutical, and medical devices sectors, supporting both commercial and research and development activities.

The areas intended for digital validation tool application include, but are not limited to:

1. **Process Validation:** (including cleaning, sterilization validation, and PPQ) Facilitates the storage of critical data related to batch release for commercial lots.
2. **Computerized System Validation and Computer Software Assurance:** Ensures compliance with regulatory standards.
3. **Facilities, Utilities, Equipment, and System Commissioning and Qualification:** Covers all stages from planning, user requirements, system classification, system risk assessment, design review, design qualification, commissioning testing, qualification testing and reports, and trace matrices. This also includes supporting project documentation and engineering deliverables, such as Design documentation, functional specifications, and system turnover packages.
4. **Analytical Equipment/Instrumentation Qualification:** Covers testing of all analytical instruments and systems, from URS to the qualification report.
5. **Analytical Methods Validation:** Manages the transfer and site testing of analytical methods between internal and external sites, such as Contract Manufacturing Organizations (CMO’s).
6. **Validation Maintenance, Requalification, Periodic Review, and Validation Planning:** Covers the maintenance of the validated state, including temperature mapping requalification of equipment, requalification of rooms (particulate and viable), and cleaning requalification of equipment (CPV).

7. Change Management:

- Manages change in all areas of validation, particularly when systems are not under routine GMP use or are outside the scope of operational change control
- Manages change for not direct impact systems, such as plant steam or waste systems
- Manages project changes for scope, budget, and schedule

8. Transport/Shipping/Packaging Validation: Ensures the qualification of shipping control measures throughout the shipping cycle to maintain product integrity

To the extent required based on the defined scope of use (e.g., GxP, Non-GxP, logbooks), the DVT should be adequately specified, verified, documented and fit for its intended use. This should be done following *ISPE GAMP® 5 (Second Edition)* [5] principles.

Once the system has been accepted and released for use, maintaining compliance and fitness for intended use throughout its operational life is critical. This is achieved through established procedures and up-to-date training that covers usage, maintenance, and management. Refer to Chapter 5 for Validation and Chapter 6 for Governance.

3.3 Data Integrity

A DVT provides opportunities to enhance data integrity in validation documentation. Ensuring data integrity throughout the data lifecycle is essential. Data integrity controls should ensure that data and records remain attributable, legible, contemporaneous, original, and accurate (ALCOA). Furthermore, data and records should also be complete, consistent, enduring and available (ALCOA+).

Tables 3.1 and 3.2 provide information of how key concepts described by ALCOA and ALCOA+ should be applied throughout the data lifecycle.

Table 3.1: ALCOA [11]

Principle	Data Expectation
Attributable	• Attributable to the person or system generating the data • Identify the person or system performing an activity that creates or modifies data • Linked to the source of the data
Legible	• Readable and permanent • Accessible throughout the data lifecycle • Original data and any subsequent modifications are not obscured
Contemporaneous	• Recorded or observed at the time the activity is performed
Original	• Original Copy is the first recording of data or a “true copy” that preserves content or meaning
Accurate	• Free from error • No editing performed without documented amendments • Conforming to truth or standard

Table 3.2: ALCOA+ [11]

Principle	Data Expectation
Complete	All data, and relevant metadata, including any repeat or re-analysis performed
Consistent	- Application of good documentation practices throughout any process - The application of date and time stamps in the expected sequence
Enduring	Recorded in a permanent, maintainable form for the retention period
Available	Available and accessible for review, audit, or inspection throughout the retention period

The requirement for data to be “*traceable*” may also be added to ALCOA+, to create ALCOA++.

Data should be traceable throughout the data lifecycle, that is, any changes to the data, to the context/metadata should be traceable, should not obscure the original information, and should be explained, if necessary. Changes should be documented as part of the metadata (e.g., audit trail) [33].

For further information on data integrity, ALCOA++, and data lifecycle refer to *ISPE GAMP® Guide: Records and Data Integrity* [11].

Table 3.3 discusses how the use of DVTs in validation can support ALCOA++ principles and data integrity and how it is also applicable to hybrid processes.

Table 3.3: How DVTs can support ALCOA++ and Data Integrity in Validation

Principle	DVT Benefit
Attributable	DVTs ensure all data is attributable through detailed audit trails for data entry, linking actions back to the user or system.
Legible	DVTs enable typed entries, which help mitigate issues associated with poor handwriting and inadequate good documentation practices. However, when attaching documentation or evidence to a DVT, it is important to ensure that the attachments are legible by specifying that they adhere to the ALCOA++ principles. This can be done as part of the “true copy” verification process.
Contemporaneous	DVTs may allow for sequencing of data entry, which will protect against non-contemporaneous execution. However, if this functionality is not available, the audit trail of the data ensures that non-contemporaneous practices are visible and can be addressed through review and approval workflows.
Original	During testing execution activities digital solutions can be used to digitally capture evidence that acceptance criteria have been met, such as material of construction certifications, drawing walkdowns, and laboratory testing data. This functionality allows other ALCOA++ principles to be satisfied (as outlined in this table). However, when transferring original records into a digital system, it is essential to ensure it is done in a controlled manner. This process should align with true copy principles, which preserve the content and meaning of the records. When the data is entered into the system, every effort should be made to ensure that: - The level of verification of the true copy is commensurate with the level of risk (i.e., if the original is easily available, or the level of regulatory oversight IQ versus PV) - Digital copies of documentation are of appropriate resolution, clear and legible

Table 3.3: How DVTs can support ALCOA++ and Data Integrity in Validation (continued)

Principle	DVT Benefit
Original (continued)	An electronic signature is available in the system at the point of entry, to attest that the digital copy has been verified against the original record People using the system are trained and made aware of the responsibility of verifying the record against the original record. This should be enforced by SOPs for good documentation practices or use of the system, or by instruction in the testing protocols/ test cases
Accurate	Another benefit of digital solutions is the enhanced accuracy of the electronic records versus paper. In a digital solution, the data “committed” is protected from change via an audit trail in line with data integrity principles. Paper processes do not have the granularity of an accurate timestamp and rely mainly on training and SOP adherence for documenting changes to records in a compliant way.
Complete	All data “committed” to a DVT has an audit trail, and no preapproved data can be deleted. Also, the audit trail has the original and any changed data.
Consistent	Sequencing of events have a date/timestamp. Therefore, when executing a sequence of test steps, a timestamp of completion at the end of each step is created.
Enduring	The data in the electronic system will also be enduring, as it will not fade, be lost, or destroyed, which are downfalls associated with paper-based systems, provided the system is properly validated and backed up.
Available	Another advantage of digital solutions is that data is available at any point in its lifecycle. This eliminates the challenges associated with paper-based records, such as locating hard copy documents, off-site storage, and retrieving records during audits.
Traceable	One of the main advantages of digital solutions is that, once data is entered into the system, it is automatically attributed to the user or system with an electronic time stamp. Additionally, any changes to the data are tracked with an audit trail that records what was changed, who made the change, and when it occurred. This helps reduce the risk of data falsification.

When required, during test execution activities, the DVT should allow for the digital maintenance of records as evidence of acceptance criteria. The ingestion and maintenance of original records in the DVT should be done in a controlled manner, ensuring that the first recorded paper/electronic record version can be discarded appropriately. This process should follow good documentation management practices.

The content and meaning of the record must be preserved, all predicate rule requirements must be met, and the ability of the regulated company to meet all regulatory and legal obligations should be maintained. The maintained records should facilitate the understanding, review, and reconstruction of the validation activities.

Once true copy verification is completed, the original records can be destroyed based on established site/corporate procedure for the destruction of the original records. These procedures should include a record of destruction, covering authorization and documentation. The process should be risk-based, taking into account the criticality of the records, the maturity of the organization, and other technical and procedural controls in place [35].

The record management process within the tool should be included in the URS as part of the implementation (Chapter 5), and the management of true copies should be included in the DVT governance (Chapter 6).

As organizations move towards more integrated digital systems and a digital ecosystem, the goal is to:

- Eliminate static paper data
- Eliminate the uploading of static electronic data by connecting systems to make these records visible in a digital validation tool
- Make dynamic data within the digital validation tool for review and approval as evidence of compliance with requirements and fitness for intended use

A true digital ecosystem should involve no copying of any data and should make original data accessible when required.

3.4 Culture Change, Organizational Change Management, and Knowledge Management

3.4.1 Review, Approval, and Execution Concepts

The adoption of DVTs within the pharmaceutical industry necessitates a cultural shift within organizations. This section outlines the key reasons why this change in culture is essential and the benefits that will result, but first, the current ways of working and the challenges and limitations they present are considered.

Validation activities using paper often occur in silos due to the nature of the tasks. For example, document review and approval that requires the physical copy of the document must be conducted on an individual basis, as each reviewer/approver must have such physical copy. Similarly, the execution of paper-based test documents must follow this approach, as altering or splitting the documents in any way could lead to missing pages or data. These paper-based processes present significant roadblocks and delays:

- The traditional model of document ownership at different milestones hinders collaboration. During draft issuance and review of a document, the author is considered the document owner, often spending significant time resolving comments where they are not the Subject Matter Expert (SME). When documents are issued for approval, the final approval typically rests with the Quality representative, implying they are responsible for the entire document's contents.
- A typical review/approval cycle for executed test documents, where multiple reviewers/approvers each have 3 to 5 days to perform their tasks, leads to significant delays.
- Reviewer/approver comments are often not shared with the entire team, resulting in the author or executor of the document needing to follow up individually with reviewers/approvers to resolve comments, further extending the review/approval process.
- Test execution activities are delayed by the need to perform tasks on a single paper copy. For instance, installation verification tests performed across multiple locations (e.g., P&ID walkdown, material of construction verification) cannot be done concurrently with paper documents, as the document cannot be split and typically require separate test occasions.

Change to current roles of reviewers/approvers:

- The current process may involve many reviewers and approvers of a document. Regulated companies should critically evaluate the roles of reviewers and approvers to optimize efficiency. Reviewers should be key SMEs, responsible for consulting with relevant SMEs to ensure document compliance and the quality of the content. The DVT should provide the user with a means to monitor and predict the average number of reviewers and approvers for each document or template; this topic is covered in more detail in Section 3.6. The DVT should also be configurable to limit these roles based on organizational needs.

- *ISPE GAMP® 5 (Second Edition)* [5] encourages the maintenance of information in a structured and flexible way rather than in traditional document format, and in such cases, a transition from a concept of “approval” to “acceptance” may be appropriate and effective. Refer to *ISPE GAMP® 5 (Second Edition)* [5], Section 28.3.5 Approvals and Acceptance.
- The role of the reviewer using a DVT changes, rather than focusing on GxP compliance of tests and attachments, reviewers can concentrate on data analysis, to ensure scientific accuracy of the data in relation to the acceptance criteria. In the framework of Pharma 4.0 in the future, reviewers will also oversee data that is integrated with automation across multiple systems.

With this culture change, it is important to note the significance of the training and education of all relevant stakeholders. ALCOA++ principles on paper also apply to the DVT; however, certain principles such as “Contemporaneous” require additional training and considerations where it applies to generation and execution of documents within the DVT.

In the paper process, the executor applies their signature or initials and the date of the execution when performing test steps. In the DVT, each entry is captured within the document audit trail where the entry is linked to the username of the executor, including a date and time stamp. This high level of transparency provides reviewers, approvers, and auditors clear visibility of actions performed, potentially highlighting issues such as out of sequence test steps or significant time gaps. This transparency can also prompt discussions with compliance teams about the validity of the tests, even if the test is not time sensitive or requires sequential execution. To address this, documentation and procedures must be adjusted accordingly to align with this transparency, and users must be trained to understand that every action is captured/recorded and cannot be undone.

The ISPE Data Integrity blog post [17] emphasizes the need for more education of users of the systems, making it clear that every action within the system is permanently captured and cannot be removed.

In tandem with this further education of users, careful consideration should be given to the generation of test document templates and test cases within the DVT. The use of instructive text in validation documents can guide executors, ensuring they have clear instructions on how test steps are to be executed (i.e., sequential: step 1, then step 2, etc., or flexible: steps can be performed in any order). Where the audit trail of the document captures the individual cell history, instructive text can ensure executors fill out fields in the proper order, maintaining data integrity.

Data should be securely and accurately recorded at the time of entry. Any subsequent editing or review should be captured while preserving the original entry. The intended use and the correspondingly applicable predicate rules determine if, where, and when regulated signatures are required.

The transition from paper-based systems to DVTs requires not just technological adjustments but also a significant cultural shift within the organization. Chapter 5 provides further guidance on fostering this culture change.

3.4.2 Collaboration and Breakdown of Silos

Engagement of key stakeholders at the appropriate stages is crucial to moving away from siloed departmental workflows. As demonstrated through the introduction of Risk-Based Commissioning and Qualification in alignment with the *ISPE Baseline® Guide: Volume 5 – Commissioning and Qualification (Second Edition)* [18], this approach offers significant benefits. For example, involving the Quality Unit in the early stages of the system risk assessment process helps to address potential risks proactively. Additionally, the participation of the end user is vital throughout the validation lifecycle. Any decisions made by the C&Q team and/or Quality Unit that may impact system operation or maintenance need to be reviewed by the end user.

According to the *ISPE Baseline® Guide: C&Q (Second Edition)* [18], there is a need to transition from viewing the Quality Unit as a stand-alone review role to an integrated member of the C&Q team, with oversight of the entire validation lifecycle for GxP impacting systems. While the Quality Unit is not expected to have expertise in all aspects of the lifecycle, SMEs should be responsible for designing and testing the systems. The Quality Unit should work in tandem with the SMEs to ensure quality requirements are met and to approve the testing plans and documents for systems with direct product impact.

The use of Digital Validation Tools enhances collaboration by enabling real-time communication between experts. These tools provide Quality oversight throughout the lifecycle of documents. The DVT can enable a more collaborative review and approval process, shifting from a single responsible person to a team-oriented document ownership model. This shift leads to more efficient processes, as outlined below:

- **Shared Responsibility for Documents** – Author is not the only owner up to the point of a document being issued for approval, all reviewers and approvers are equally responsible. This eliminates the notion that the Quality Unit is the ultimate authority. Document ownership is clearly defined, preserving accountability and attributability at every stage.
- **Collaborative review** – rather than the author being responsible for the resolution of all comments, all reviewers share the responsibility for resolving comments. This collaborative approach improves the efficiency and effectiveness of the review cycle, enhancing document quality.
- **Concurrent Approvals within the DVT** – meaning that multiple approvers can review the same document simultaneously, streamlining the approval cycle.
- **Live execution with the DVT** will enable multiple executors to perform tests simultaneously within the system without impacting each other, improving the efficiency of the execution cycle.
- **Live execution within the DVT** enables reviewers to perform their reviews in real-time; this in turn leads to a reduction in test discrepancies and repeat testing. A truly effective DVT should provide reviewers/approvers with the ability to add comments on executed tests in real-time without the requirement for the document to be issued for final review or the need for the reviewer/approver to leave the system to provide their comments to the executor.
- The use of DVTs offers several advantages over traditional paper-based execution practices, streamlining processes and reducing inefficiencies.
- In paper-based systems, blank spaces in executed document must be marked as “Not Applicable.” DVTs eliminate this requirement by maintaining a full audit trail of all executed entries. Once the document is approved, the DVT locks any incomplete cells, preventing the need to mark them as “Not Applicable.”
- In paper-based processes, executors are required to annotate and paginate any attachments/evidence in line with good documentation practices to ensure proper attribution to the correct test. DVTs simplify this process by allowing executors to attach relevant attachments/evidence directly to the specific test step being executed. These attachments are captured within the document’s audit trail, eliminating the need for manual annotation, pagination, and the separate attachment section typically required in paper-based documentation. More details on this process are covered in Chapter 6 – Digital Validation Tool Governance.

3.4.3 Knowledge Management (KM)

The incorporation of KM in pharmaceutical organizations has been slower than in other industries. Beyond the specific arena of pharmaceutical quality, KM is increasingly recognized as a stand-alone discipline that focuses on how organizations create, manage, and utilize knowledge. This is particularly relevant in today’s business environment, where knowledge workers dominate various sectors and complex digital systems are widely used to store the growing body of knowledge generated across industries. ISO 30401 Knowledge Management Systems – Requirements [19] was published in 2018 with the stated objective to:

“...support organizations to develop a management system that effectively promotes and enables value-creation through knowledge.” – refer to the *ISPE GPG: Knowledge Management in the Pharmaceutical Industry* [34], Section 1.1.

The application of KM can empower key SMEs in all areas of the organization. For example, it enables SMEs to present in-audit scenarios of the relevant documentation and information requested by auditors with the same level of confidence as audits conducted prior to the introduction of DVTs. Refer to Chapter 7 Regulatory Considerations/Impact, which covers internal audits and regulatory techniques and challenges.

Embracing cultural change and breaking down organizational silos facilitated by effective Organizational change management and KM help create a more efficient workforce of knowledgeable SMEs. This cultural shift will expedite the seamless adoption and effective use of DVTs, driving efficiency and improving overall outcomes.

3.5 Data Accessibility

A core concept of any DVT is ensuring the accessibility and availability of data related to Key Performance Indicators (KPIs) and metrics. This is essential for implementing and maintaining an effective continual improvement program.

The data available in a digital validation tool should support the generation of metrics and KPIs used to monitor and control quality, safety, and business processes. Organizations can leverage this data for Lean Six Sigma process optimization, utilizing the DMAIC (Define, Measure, Analyze, Improve, and Control) methodology, explained as follows [20]:

1. **Define** the problem, improvement opportunities, project goals, and customer (both internal and external) requirements
2. **Measure** process performance
3. **Analyze** the process to identify root causes of variation or poor performance (defects)
4. **Improve** process performance by addressing and eliminating the root causes
5. **Control** the improved process to sustain performance over time

For more information on DMAIC, visit: <https://asq.org/quality-resources/dmaic> [20]

The data should be categorized into two key areas: Quality and Business. The Quality aspect is critical for demonstrating process compliance and ensuring product quality and patient safety. The Business aspect focuses on analyzing and improving the efficiency of the business processes of validation to reduce time, enhance planning, and provide visibility for ongoing measurement and adjustments as needed.

3.5.1 Quality Metrics/KPIs

The following list includes potential Quality Metrics/KPIs but is not limited to:

- Improvement in product quality or reliability
- Increase in customer satisfaction
- Reduction in product recalls or regulatory action
- Document approval dates (e.g., date of qualification/validation)

- Test failures/fields with “fail” status
- Tracking of the Validation Status of Equipment/facility/utility/Automation/computerized/analytical instrument Systems (e.g., a traffic light system)
- Requalification /Periodic Review Scheduling (cleaning, sterilization, temperature control units, automation systems, continuous process verification)
- Validation Discrepancy/Non-Conformity/Incident/exception Monitoring, categorization, root cause, and status (Open/Closed)
- Empty or incomplete fields
- Fields that have been changed
- Document traceability and status dashboarding
- Requirements and Risk Traceability

3.5.2 Business Metrics/Key Performance Indicators (KPIs)

Business metrics/KPIs can be applied across various documents, including plans, reports, execution protocols (pre and post review and approval), qualification/validation discrepancies, non-conformities, and incidents.

The following list includes potential business metrics/KPIs but is not limited to:

- Time to market for validated products
- Track progress by department/role
- Earned value management by measuring progress compared to target goals
- Number of review cycles

Draft Stage:

- Date drafting document commenced
- Percentage content increase since drafting commenced (number of tests, number of fields)
- Percentage rework between versions

Review Stage:

- Date document sent into review
- Date Department 1 completed review
- Date Department 2 completed review
- Date Department 3 completed review
- Review complete date

- Time taken to review (by person, by department), actual time spent interacting in the document
- Number of comments per reviewer
- Actual reviewer completion time, (by person, by department)
- Number of comments per department
- Number of comments per document
- Comment categorization – Value Add/Non-Value add

Approval Stage:

- Date document sent for approval
- Date Department 1 completed approval
- Date Department 2 completed approval
- Date Department 3 completed approval
- Approval date
- Approval iterations – track rejections

Execution:

- Date document commenced execution of first field
- Date of final executed field
- Date executed document issued for review

It is important to identify the key metrics to measure using the Digital Validation Tool. These metrics should be defined as user requirements for the DVT (refer to Chapter 5). Additionally, they should be incorporated into the governance framework of the Digital Validation tool to clearly define what is being measured and the parameters for measurement.

3.6 Digital Validation Integration

Integrations with a DVT offers significant benefits for regulated companies, primarily through time and effort savings, increased efficiency, and a reduction in both error potential and compliance risks. Properly validated integrations ensure consistency in the results across applications by preserving both record content and meaning. Each integration should be thoroughly validated, providing evidence of repeatability, reproducibility, and robustness while also ensuring that data transfer adheres to ALCOA++ principles.

It is important to note that data transfer may not be necessary in all cases. Instead, an integration may involve a call-out or path to the data source, such as integration with a data management system. This section focuses on the concept of integrating other software with a DVT, rather than the specifics for implementation.

Integrations may occur directly or require the use of an Application Programming Interface (API) with middleware, depending on the software being integrated. Specific integration requirements should be outlined within the DVT Requirements Specification. All integrations must be properly validated.

Where the two systems interact, user access considerations must be carefully evaluated. If the entire document's data is not transferred (i.e., using hyperlinks instead of transferring embedded data or pdfs), users may encounter issues accessing all aspects of the documents, including attached evidence, source documentation, cross-references, etc. To avoid such problems, it is recommended to embed the data directly into the protocol as part of the integration, rather than relying on hyperlinks or cross-references. Alternatively, the integration may allow access to dynamic data within the secondary system where users can only view the specific data being requested by the DVT. This approach would enable users to "peer" into the secondary system without accessing it directly.

As with any integration, careful considerations should be given before proceeding with implementation. For more detailed guidance on integration considerations, refer to Sections 5.3 and 5.8.

3.6.1 Document Management System (DMS)

A user may integrate their DVT with a Document Management System (DMS) to transfer documentation between the two systems. In some cases, the DVT may not be the chosen or validated document repository for the regulated company, and all approved documents are to be stored within the DMS. In such instances, an integration can be set up to push documentation from the DVT to the DMS for storage.

This integration can be event triggered (e.g., when the document is approved within the DVT, the integration is triggered and sends a copy of the document to the DMS) or based on a transfer frequency (i.e., once a week, the integration recognizes newly approved documents within the DVT and transfers them to the DMS). Regardless of the approach, considerations must be made regarding how approval of the document is handled within the DMS.

With a validated integration, ideally there should be no need for approvals within the DMS; however, the DMS may not support capturing external approvals or bypassing the approval process. In such cases, the following options should be considered:

- Single approval within the DMS – rather than requiring all original signatures, a single approval can confirm the transfer of the document
- Single-stage approval process – instead of a multi-stage approval process, a single-stage process can incorporate the original signature capacities in one step

In any validated integration, the meaning of an approval or signature within the DMS may differ from its meaning within the DVT. It is important to clearly define the significance of a signature in each system.

If the DVT is chosen as the validated document repository for documents created within the DVT, integration with the DMS may be unnecessary. However, integration may still be utilized to reference and link information from the DMS into the DVT documentation or vice versa, if the systems do not already support hyperlinking to external sources. Links to the DMS can be either static or dynamic.

- Static link – a snapshot of the reference as it appears in the current revision
- Dynamic link – a live link that updates as the document itself updates

3.6.2 Scheduling Tools

Integration with the Regulated Company's scheduling tool may be established. This integration is typically a one-way pull, where the scheduling tool requests document status information from the DVT and then updates its line items accordingly. This integration supports project oversight by enabling the scheduling tool to adjust timelines for document drafting, execution, or system release as progress is made within the DVT.

While many factors influence the timeline for system release, lifecycle documentation is a critical dependency, often under the Regulated Company's control. This integration ensures that the schedule remains up-to-date, providing an accurate reflection of project status (on time, early, or delayed).

3.6.3 Change Management or Deviation/CAPA Management Systems

If deviations are not managed within the DVT, an integration may be introduced to initiate a deviation within the Deviation Management system when specific parameters are met (e.g., when the protocol has a "Deviation required?" checkbox activity ticked). With proper integration, data from the discrepancy within the DVT can be pulled into the deviation management system, auto-populating specified fields to save user time and enhance consistency and efficiency.

Once the deviation has reached a specified milestone, and if further testing is required, the integration could work in reverse, from the Deviation management system to the DVT, to create new documentation as required by the deviation assessment, again transferring only the specified data.

Additionally, an ability to preview the status of a deviation. This preview could provide insights such as the stage of the deviation or whether the protocol for additional testing has been approved, allowing the deviation to proceed.

3.6.4 User Management and Learning Management Systems

Integrations can be established between the User Management System, Learning Management System (LMS), and a ticketing system (optional) to automate onboarding and offboarding processes. When an access ticket is submitted, the integration may pull applicable information from the User Management System and the LMS to verify the user's onboarding status, role/responsibilities, and completion of required training. The integration will then auto-create the user account within the DVT, granting the appropriate roles and responsibilities. This would be a triggered event, initiated by the ticketing request and is likely the most common scenario since approval for the ticketing process may be required before the integration is triggered.

In the absence of a ticketing system, an API could detect a new user account within the User Management System, verify the applicable training via the LMS, and assign appropriate access based on the user's role within the User Management System and completed training. This approach would most likely operate on a query-based schedule (e.g., daily), where the DVT scans for new users, and once identified, verifies the user's training and assigns appropriate access. This approach:

- Is highly dependent on the configuration and setup of both the User Management System and LMS
- Assumes ticketing approvals are not required
- Requires clear guidelines around onboarding, personnel roles and role assignments within the DVT
- Requires control within the LMS over who can train on the relevant DVT documentation
- Places additional demand on the integration, as it will identify all new users in the User Management System, regardless of whether they require access to the DVT

This approach may be optimized by allowing the System Admin to manually enter the unique User ID, prompting the integration to scrape both the User Management System and LMS for the required information to create the user's account within the DVT.

The integration could also monitor the user's training status, ensuring they remain current not only on DVT-specific training but also on training relevant to their role (e.g., GxP, Validation Programs). This would allow the system to link specific tests and procedures to the user's training compliance, ensuring they can only execute tasks based on completed training. Users would be notified if they fall out of compliance (e.g., temperature mapping, change management, or specific equipment usage). This integration would eliminate the need for training signature forms within Validation documentation.

In addition to streamlining onboarding, the offboarding process can be automated. When a user is offboarded within the User Management System, the DVT can be signaled to deactivate their corresponding account. This ensures proper access restrictions are in place, helping to maintain security and compliance across multiple systems.

3.6.5 Computerized Maintenance Management System (CMMS)/Asset Management System

An integration can be used to auto-populate protocols within the DVT with information (asset metadata) from the Regulated Company's Asset Management or CMMS system. Information such as Asset Tags, Make/Model, Serial Number, calibration status (even attaching the calibration certificate), etc., can be automatically populated. A validated integration of this type not only saves users considerable time and effort but also ensures the accuracy of the recorded information. Additionally, the system could initiate validation studies based on the data stored in the CMMS, such as triggering scheduled processes for maintaining the validated state or periodic review activities.

The DVT can also report information back to the CMMS or Asset Management System, updating the asset status or attributes based on findings during qualification activities. This could include updating attributes that were not specified or were found to differ from expectations but were accepted. The acceptance and approval of the validation effort may drive updates to the information found within the Asset Management System.

3.6.6 Data Historians/Data Lake

Integrating with the user's various information databases can save time, reduce effort, and ensure consistency while eliminating transcription errors. During protocol testing, data can be directly pulled from the data storage system into the protocol, rather than being manually entered. This integration can be active in real-time during execution or used post-testing to retrieve data from specified assets, time points, or samples.

When the integration is running (whether toggled manually or automatically activated when accessing the DVT), real-time, single point or continuous data can be retrieved. The Asset Tag is recorded either during protocol generation or manually recorded by the user during execution. This information is essential to ensure that the correct data is retrieved.

- Scenario #1: Single-point pH reading for a buffer sample

In this scenario, the user either initiates the recording through an integration panel, or the integration automatically pulls the data when the recorded value cell is selected. The current pH reading is retrieved from the data storage system and imported into the DVT.

- Scenario #2: Temperature trend of a bioreactor

For continuous data, such as the temperature trend of a bioreactor, data is brought in and attached as a datasheet (e.g., .xls, .csv). The historian system is set up to record data from the tag at specific intervals, and this data is reflected within the attached datasheet, as seen in the historian interface.

In these scenarios, the tool may not require transcription of the entire dataset, but rather it provides a path to the original data, a process known as using *dynamic data*. The integration can pull the values of critical timepoints or calculated values, while the entire dataset is simply linked between the two applications.

3.6.7 Laboratory Information Management Systems (LIMS)

Similar to the functions of data historians (as discussed above), the integration of a LIMS with the DVT is specific to the laboratory, its equipment, and the samples tested. This integration ensures that laboratory data is efficiently and accurately incorporated into the validation process.

3.6.8 Temperature Validation System

As with the data historian and LIMS integrations mentioned above, the integration with a temperature validation system focuses on transferring large volumes of data from a stored location (in this case, a stand-alone piece of equipment) into the test document. Each run within these validation systems is assigned a unique identification number, which is entered into the document or integration. The corresponding test protocol includes the data printout (e.g., .xls, .csv).

To optimize this process, a standardized mapping test script should be created and used within the integration setup. Specified data points could then be automatically populated, minimizing user involvement. The datasheet would be attached to the specified location, and values such as the run's Minimum, Maximum, and Average could be automatically populated into the corresponding fields. With clearly defined acceptance criteria, the system could identify and notify the user of any failures.

Given that these systems are often used for various validation activities, the integration must account for different setups, which could be either user-selected or determined based on the test script used. For example, activities such as chamber mapping or lethality could be selectable within the integration to ensure that the proper data is collected and populated within the specified locations.

As the systems are not typically directly nor fully connected to remote data storage systems, the necessary data needs to be manually transcribed into the DVT. This is considered static data (as opposed to dynamic data). This approach can be applied to other types of stand-alone equipment.

3.6.9 Engineering Information Management Systems (EIMS)

Integration with EIMS can streamline the capture and management of Engineering Specifications within the system's lifecycle. Examples of how this integration could be beneficial include:

- Extracting Critical Process Parameters (CPPs) and specifications to be directly included in requirements documents or test protocols as acceptance criteria.
- Linking documents, drawings, specifications, and reports to the Design Qualification (DQ) as supporting references.
- Enabling comprehensive traceability of requirements across various documents, both within and outside of the DVT, to ensure all critical and non-critical requirements have been met.

Additionally, integration with the EIMS allows for assessing the impact of process changes across the current document lifecycle or during New Product Introduction (NPI). For example, it could help determine whether additional testing is required, or if the current validated state remains sufficient.

3.6.10 Distributed/Process Control Systems (D/PCS) and Manufacturing Execution System/Manufacturing Batch Records (MES/MBR)

Integrating the DVT with D/PCS allows for the capture of real-time events during testing, such as Sequence of Operations. As a sequence progresses, data can be sent to the Test Protocol, automatically checking off each step as it occurs, while also signing and dating each action as completed. If predefined acceptable parameters are outlined, the system can also verify whether these parameters were met during the process, alerting the user to any out of specification occurrences.

In another scenario, without predefined steps (unscripted testing), actions performed during testing could be automatically recorded within the test script, providing a full record of events.

This integration is particularly valuable during Process Validation (PV), where large volumes of data is gathered and reviewed for release. By integrating with both the D/PCS and MES/MBR systems, all necessary data can be directly accessed, eliminating the need for manual data transcription.

Additionally, using advanced integrations with AI and dynamic data, the data recording process may be unnecessary. The data could be verified within the data source and accepted or rejected directly within the validation exercise. Using dynamic data, the source remains available for further investigation as deemed necessary or requested.

Process Validation is often supported by Batch Records, and if the test lots are to be released then these Batch Records will be required regardless. Integrating the DVT with the MES/MBR Systems can help confirm the information stored within the records, supporting the protocol and the decision to accept or reject the release of the PV lots.

Connecting the DVT with both the D/PCS and MES/MBR systems provides a comprehensive suite of PV tools, streamlining and automating many critical validation processes.

3.6.11 Electronic Logbooks

Integration with electronic logbooks supports system calibration and standardization tasks. Many analytical instruments require daily standardization, which is recorded in a logbook. By integrating the DVT with the electronic logbook, the system can verify the current calibration status of the equipment to ensure it is ready for use.

On the other hand, electronic logbooks are often used to record system usage. When an analytical instrument's tag is recorded within the DVT, the system can automatically send the information to the electronic logbook to create a usage entry. This entry would include necessary details such as the operator's name, timestamp, recorded value, and other relevant data.

3.6.12 Materials Requirements Planning (MRP)/Enterprise Resource Planning (ERP)

Integrating with MRP/ERP Systems can help automate the process of material requests based on testing requirements. For example, an integration could automatically request materials (e.g., disposable single-use items) based on the testing to be performed and inform the user about availability of materials for planning and scheduling. By scanning materials during testing, the test document could auto-populate the information required prior to execution and provide notifications about material expiration to ensure materials are used within their valid periods.

3.6.13 Robotic Process Automation (RPA)/Bots

The DVT should be accessible to RPA systems and its data available to RPA/Bots for tracking, trending, and presentation. Refer to Section 3.5 Data Accessibility for further discussion.

3.6.14 Integrations Between Two Separate DVTs

Integrating two DVTs can be beneficial for improving supplier/customer relationships. When two DVTs communicate easily, it simplifies and streamlines the documentation handover process, facilitating smoother Validation activities.

Some examples include:

- One DVT at a CMO connected to another DVT at the manufacturer
- One DVT connected with another DVT within the same company in a different department or division
- Vendor/Supplier integrating one DVT with a customer's DVT
- Enabling integration between DVTs with different versions

3.6.15 Integrations between a DVT and Automation Testing Tools

Automated testing tools can facilitate efficient change management and regression testing by verifying critical functionality and ensuring that regulated data is not adversely impacted after application upgrades. Performance metrics generated from automated tests help confirm that solutions remain fit for intended use, while ensuring patient safety, product quality, and data integrity throughout the changes [10].

Integration with a DVT and an automated tool improves efficiency in the execution and planning of activities, eliminating reliance on manual user inputs once the system is configured. For example, the DVT can automatically notify the external tool to initiate testing, and the tool can provide auto-confirmation upon completion of the test script. Additionally, it can pull test result data, if required, and gather information from repositories to fulfill installation testing requirements.

4 Crafting a Compelling Business Case for a Digital Validation Tool

4.1 Introduction

Embracing A DVT is not merely a technological shift; it is a strategic initiative that aligns with corporate objectives, enhances operational efficiency, and strengthens compliance in support of patient safety and product quality. By clearly addressing the challenges of traditional validation processes, showcasing the compliance advantages, and presenting well-defined return on investment, organizations can build a compelling case for adopting a DVT in support of patient safety and product quality.

The adoption of a DVT also aligns with corporate initiatives. This business case outlines the justification for the investment by defining key challenges, integrating the benefits, assessing risks, intersecting with broader corporate goals, showcasing the technology, analyzing stakeholder impact, and demonstrating how a DVT ensures compliance. This Guide explores the essential factors to consider when crafting a compelling business case for the adoption of a DVT.

Current State and the Need for Change

Companies with manual, paper-based process workflows often face inefficiencies, higher costs, and increased compliance risks. To effectively make the case for change, it is important to thoroughly understand the current processes, not only by mapping them but also by understanding the stakeholders and how they are impacted by an inefficient paper-based system. Questions to consider are as follows:

- Does our system cause bottlenecks?
- What inefficiencies exist in the current workflow?
- What impact do these inefficiencies have on project resources, timelines, and budgets?

These initial questions will help uncover areas where a DVT can resolve challenges. By automating critical processes, a digital system minimizes errors and reduces operational delays, thus improving the efficiency and reliability of the validation process.

Benefits of Digitizing the Paper-based Process

A strong business case highlights the advantages of transitioning from paper-based processes to a DVT. There are numerous benefits to digitizing a paper-based process.

A major benefit of moving from paper-based processes is the opportunity to strengthen data integrity through the use of validated interfaces for data capture, computer-generated audit trails, and attributable operator actions.

Other benefits include improving workflow efficiency, reducing the overall cost of validation within a project or within operations, in the case of periodic reviews, maintaining the qualified state, as well as improving compliance

These advantages are just some of the key benefits as there are other use cases based on the organization (e.g., CxMO). By carefully mapping out workflows and identifying inefficiencies and costs, it becomes easier to quantify the return on investment (ROI) for stakeholders. Additionally, qualitative benefits such as improved customer trust due to higher levels of quality and compliance, and greater employee satisfaction from a streamlined process, can further strengthen the business case.

Alignment with Short and Long-Term Strategic Corporate Goals

When writing the business case, it is also important to align the need for a digital solution with both short and long-term corporate goals.

- Understand what those goals are and include that integration with the digital solution.
- Consider how the digital solution will support innovation, scalability, adaptability, market share growth, sustainability or other strategic goals from an organization.
- Demonstrate to stakeholders that the investment does not merely ensure short-term benefits, but that the digital solution will provide enduring benefits long after the initial investment.

Gauging the Risks

There are many risks associated with maintaining the status quo. It is important to assess these risks and include them in the business case. Potential risks include:

- Increased cost for capital expenditures and operations
- Compliance issues due to outdated processes
- Ongoing inefficiencies in a paper-based process
- Customer dissatisfaction with the quality of the documents
- Missed business expansion opportunities

By highlighting these risks, the business case underscores the importance and urgency for adopting a DVT. Presenting the digital solution as a risk mitigation strategy demonstrates its value to stakeholders.

4.1.1 Defining the Challenges

4.1.1.1 Identifying the Challenges and Opportunities

The first step in creating any business case is identifying challenges and opportunities. Listening to the voice of the customer can be a place to start. By understanding what causes the greatest frustration in a paper-based validation process, a compelling problem statement that resonates with decision-makers who will finance the project can be crafted. Common challenges and opportunities in validation projects include extended timelines, rising costs, and the difficulty of managing paper-based documents as they travel from one reviewer to the next.

Additionally, providing status updates on these documents can become cumbersome. These inefficiencies can delay the start of manufacturing, potentially costing the company large amounts of money, not to mention the impact on patients waiting for the products to become available.

4.1.1.2 Reduction in Good Documentation Practice Errors

A key element of pharmaceutical validation is ensuring adherence to good documentation practices. In a paper-based system, maintaining good documentation practice standards can be challenging. Errors can go unnoticed, leading to compliance issues and audit findings. By utilizing a DVT, good documentation practice errors can be greatly reduced or eliminated, accelerating the approval process. With digital time stamps, it becomes immediately apparent when a test is conducted, and any missing sections can be easily identified. This increased transparency improves compliance, reduces the risk of errors, and streamlines the overall review process. This means that reviews are more focused on data and results rather than identifying types of good documentation practices errors, enhancing overall efficiency.

4.1.1.3 Support for Remote Work

The COVID-19 pandemic accelerated the shift toward remote work across various industries, including pharmaceutical manufacturing. A DVT aligns seamlessly with this shift, by providing robust support for remote collaboration. The adoption of a DVT reduces the need for vendor travel and enables teams to collaborate and review documents across multiple time zones, thereby reducing travel costs while improving flexibility and adaptability.

4.1.1.4 Data Tracking and Issue Resolution

In a traditional paper-based environment, tracking issues and resolutions can be challenging. Metrics related to issue resolution times are often undocumented, making it difficult to measure and improve processes effectively. A DVT empowers organizations to track issues efficiently by recording the time it takes to raise and resolve issues, eliminating the need for separate manual lists. This data-driven approach enhances accountability, facilitates continual improvement, and provides valuable insights for process optimization.

4.1.1.5 Validation Paper Library

Another challenge in paper-based validation processes is managing the storage of documents. Physical copies of validation documents can be resource-intensive and space-consuming. A DVT eliminates the need for physical storage, allowing organizations to maintain a digital library of validation documents. This reduces storage costs and enhances document accessibility, ensuring that records are readily available and secure.

4.2 Highlight the Benefits

4.2.1 Reduction in Data Integrity Risks

Data integrity is paramount in the pharmaceutical industry, as a compromise can result in severe regulatory consequences. Traditional paper-based processes are inherently susceptible to data integrity risks, including human errors and document tampering. A DVT mitigates these risks by providing robust data security and audit trails. Electronic records are less prone to manipulation, and access control measures ensure data integrity is maintained throughout the validation process.

4.2.2 Streamlined Audits

Audits play a vital role in pharmaceutical manufacturing. A DVT streamlines the audit process by granting auditors access to electronic records. This expedites audit requests, as electronic documents are easily searchable and retrievable.

4.2.3 Consistent Standardization

Consistency and standardization are essential aspects of compliance in pharmaceutical manufacturing. Regulatory agencies require organizations to maintain standardized processes and documentation. A DVT enforces consistency and standardization by design, ensuring that all validation processes adhere to predefined standards. This reduces the risk of compliance issues and minimizes the likelihood of audit findings.

4.2.4 Historical Data

Pharmaceutical companies often struggle with the lack of historical data in paper-based systems, which can hinder decision-making and risk assessment. A DVT addresses this gap by providing easy access to historical data, including audit findings and validation records. Having this information readily available enhances the ability to make informed decisions and enables a thorough assessment of the organization's compliance history.

4.3 Cost/Benefits/Overall Added Value Cost-Benefit Analysis

Crafting a compelling business case for a DVT requires a comprehensive cost-benefit analysis that evaluates both expenses and potential advantages. This analysis should include the typical categories such as software licenses, hardware, training, and ongoing maintenance as well as the costs associated with deploying the solution across a broad audience. These expenditures should be weighed against the potential benefits, including increased efficiency, cost savings compared to traditional paper-based methods, improved productivity, enhanced customer satisfaction, and stronger compliance.

To strengthen the argument, the analysis should include both quantitative metrics, such as the expected Return on Investment (ROI) for capital and operational expenditures, as well as qualitative benefits such as improved customer satisfaction. By presenting a balanced view of costs and benefits, decision-makers can better understand the overall added value that the proposed solution brings.

4.3.1 Efficiency Gains for Manufacturing Systems

One of the key benefits of a DVT is its ability to optimize production systems. By reducing equipment testing times, organizations can enhance production capacity and overall efficiency. With a DVT, the end-to-end time is reduced since there is less time spent managing the location of documents. They are stored within the DVT, allowing them to be reviewed concurrently and approved per the organization's workflow. This reduction in document handling, review, and approval time accelerates system release and decreases the number of hours spent performing these tasks, allowing personnel to focus on more critical activities.

4.3.2 Data Reusability

A DVT simplifies access to previously developed documents such as specifications, assessments, or testing protocols. The information can be reused as part of a knowledge management strategy, with the utilization of previous risk assessments and proven test cases. Additionally, testing completed on-site may be leveraged where justified. Digitization can also serve as a high-quality source for AI/ML learning models, further enhancing the utilization of the data.

4.3.3 Paper Reduction

The adoption of a DVT leads to a significant reduction in paper usage. This includes eliminating the need for printing, on-site or off-site paper storage, and the time-consuming process of requesting paper documents from storage. This reduction in paper consumption also aligns with sustainability initiatives, lowering the environmental footprint of the organization.

4.3.4 Sustainability

Highlighting sustainability advantages in the business case can further strengthen the argument. The reduction in paper usage and storage not only translates to cost savings but also to significant environmental benefits. By adopting a DVT, organizations can demonstrate their commitment to eco-friendly practices.

A well-structured DVT business case must not only incorporate these benefits but also demonstrate how a DVT supports broader corporate initiatives, streamlines processes, and optimizes resource allocation. This transition to a digitally enabled validation system paves the way for achieving Pharma 4.0's vision of knowledge-driven innovation, where manufacturing is smarter, more agile, and better equipped to meet future demands.

The adoption of a DVT represents a forward-thinking investment in the future of pharmaceutical manufacturing. By focusing on compliance, quality, and efficiency while aligning with corporate goals, organizations can confidently move toward a more innovative, sustainable, and patient-centric approach to manufacturing excellence.

Table 4.1: Benefits – (1 of 2): Capital Projects, Changes or Requalification

Area	Benefit	Explanation of Benefit
Preparation	Reduced time preparing documents (Include pre-approval review and approval times)	Preparation time is reduced as: - The time taken to review and approve documents is reduced through parallel electronic review (shared interactive commenting) and approval possible (including remote approval). - Auto-populating of certain templates (e.g., RTM and Reports) will reduce the time taken to prepare and review documents. - Standard templates with standard test cases will reduce preparation and review time.
Execution	Reduced execution time (include time preparing attachments, issue management)	Execution time is reduced as: - Attaching test evidence will be quicker. - Issues raised will be addressed sooner (as incidents are generated and resolved within the system). - Repeat test cases (required where a test fails) can be created immediately.
Execution	Cost reduction during execution (e.g., through reduced travel)	Execution costs are reduced through a reduction in travel (e.g., Supplier travel reduced or Quality representative may not be required to be present for FAT, due to option of remote approval).
Approval	Reduced time reviewing and approving documents post execution	Approval time will be reduced as: - With parallel review and approval possible (including remote approval) the time taken to approve documents will be reduced. - Testing can be reviewed as it progresses as the reviewer will have access to the live testing data as it's recorded (rather than waiting for a completed protocol). - The ability to access exhibits and referenced documents via hyperlink will improve the quality of the review while reducing the approval time.

Table 4.2: Benefits – (2 of 2): Capital Projects, Changes or Requalification

Area	Benefit	Explanation of Benefit
Data Integrity	Time saved not assessing/discussing data integrity issues or repeating work due to data integrity concerns	With legible entries which are attributable and time stamped data discussions on data integrity and any requirement to repeat tests will be eliminated.
Data Integrity	Improved data integrity	Data integrity is improved due to improved legibility, with data clearly attributable to an individual.
Management/Reporting	Reduced time tracking and reporting the status of documents	A saving is realized as: • Status reports can be generated by the system • No looking for documents as every document is accessible by everyone.
Handling	Reduced time handling documents (include scanning and archiving)	100% saving on document handling is achieved: • No paper approved controlled copies of test protocols need to be produced, distributed and controlled. • Completed executed documents do not need to be scanned and checked. • Paper copies of completed protocols do not need to be archived.
Resources	Less paper is used	100% saving on paper is achieved as paper is not used.
Other	Other benefits or limitations of using digital validation	• Incidents are automatically populated in reports so they are not missed or left open (unresolved) leading to improved compliance. • Site no longer managing templates in a paper-based system.

5 Digital Validation Tool Implementation

5.1 Introduction

This chapter outlines essential steps and considerations for the implementation of a Digital Validation Tool (DVT), which is typically delivered as a Software as a Service (SaaS) solution. The goal is not to prescribe a one-size-fits-all approach but to offer general guidance on key elements and factors that organizations should consider and address throughout the stages of the implementation process. The principles and lifecycle approach described in *ISPE GAMP® 5 (Second Edition)* [5] should be applied.

Critical to the success of DVT implementation are a well-conceived strategy and organizational readiness for digital transformation. These elements play a pivotal role in achieving successful implementation and realizing a significant return on investment. Equally important is cultivating a workplace culture that supports digital transformation, adeptly managing change, and securing senior-level support for strategic deployment of digital technologies. This recognition is vital in today's business landscape, which is increasingly dominated by digital advancements.

5.2 Creating Culture for Digitalization

This section emphasizes the importance of establishing a digital mindset across all levels of an organization. Such a mindset is fundamental for driving innovation, adaptability, and navigating the evolving landscape of digital technology and its applications in various business processes, including validation.

Emphasis is also placed on developing an organizational environment that is not only receptive to, but also enthusiastic about, digital transformation. This involves fostering an environment of continual learning, openness to new technologies, and the willingness to explore new methodologies.

The goal is to offer a blueprint for organizations to create a dynamic and resilient culture, one that thrives on the opportunities presented by a DVT. By embracing these principles, organizations can position themselves to leverage the full spectrum of benefits offered by digital technologies, leading to enhanced efficiency, improved accuracy, and a competitive edge in their respective industries.

5.2.1 Establishment of a Digital Mindset as a Foundation for Innovation and Adaptability

A digital-first mindset is essential for driving innovation and ensuring adaptability within an organization. It enables organizations to effectively navigate the evolving landscape of digital technologies, and integrate them into various business processes, including validation. This cultural shift is fundamental to staying competitive and relevant in the modern business environment. It also enables initiatives like Pharma 4.0™ [29], and drives moving towards a seamless flow of data within an organization, leading to data integrity by design and improved operational efficiency.

5.2.2 Building a Supportive Environment for Digital Transformation

Building a culture that supports digital transformation involves promoting continual learning and openness to new technologies. This includes fostering an atmosphere where innovative thinking and experimentation are encouraged. Aligning digital initiatives with an organization's strategic goals ensures that the digital transformation efforts are coherent and integrated with the broader business objectives. Additionally, providing opportunities for personnel to improve skills in the digital arena facilitates adoption and enables the development of a digitally literate workforce. This supportive environment is key to successfully integrating a DVT into core operations, enhancing efficiency, accuracy, and competitive advantage.

5.2.3 Maturity Assessment

A maturity assessment evaluates an organization's current state of readiness, capabilities, and processes for implementing a DVT. It involves assessing various dimensions, including technology infrastructure, processes, people, and compliance. The primary objectives of the assessment are:

- **Understand the Current State:** Assess the organization's existing practices, technologies, and methodologies related to validation processes. This provides a baseline understanding of the organization's readiness for digital transformation.
- **Identify Gaps and Opportunities:** Pinpoint areas of inefficiencies or opportunities for improvement. This includes assessing the maturity of existing validation processes, technology infrastructure, and the overall digital landscape within the organization.
- **Define Future State:** Establish a vision for the desired future state of a DVT within the organization. This involves setting goals, defining key performance indicators (KPIs), and identifying steps to advance maturity.

The benefits of incorporating a digital maturity assessment into the digital validation strategy are immense. Understanding the current state of an organization's digital maturity enables the development of targeted approaches to foster a digital culture that supports the successful implementation of a DVT.

Incorporating a digital maturity model assessment into the digital validation strategy is essential for ensuring that validation processes are aligned with broader organizational goals and digital transformation efforts. The following five-step approach to tailoring a Digital Validation Maturity Model should be considered:

1. Develop a Comprehensive Understanding of Digital Maturity Models Specific to Digital Validation

Start by introducing the concept of digital maturity models within the context of validation systems. It is important to educate personnel on the difference between general organizational digital maturity and the specific maturity of a DVT. This distinction ensures that efforts are focused on enhancing the digital capabilities most relevant to validation processes. Compile resources and conduct training sessions to explain various digital maturity models and their applicability to digital validation processes. Emphasize the importance of a model that can be applied generically across different organizational validation contexts.

2. Define the Scope of the Maturity Assessment

Clearly outline the scope of the digital maturity assessment to focus on the validation system. This involves specifying that the assessment will only evaluate the digital validation processes and not the entire organization or the plant's digital maturity. If multiple validation processes exist throughout the organization, benchmark them individually to identify where resources should be focused for deployment to enhance ROI. Create documentation that outlines the scope of the maturity assessment, including its objectives, the areas it will encompass, and how it is different from broader organizational maturity assessments. Ensure this documentation is accessible to all stakeholders involved in the validation process.

3. Align the Digital Validation Process's Maturity with Organizational Goals

Integrate the findings of the digital validation maturity assessment with the organization's overall digital maturity and transformation roadmap. This step ensures that the digital validation strategy is not developed in isolation but as an integral component of the digital transformation strategy. Facilitate workshops or meetings between the validation team and organizational leadership to discuss and align the digital validation strategy with the organization's digital transformation goals. Develop a plan that integrates these objectives, emphasizing how the validation process supports broader digital transformation efforts.

4. Incorporate the Digital Validation Maturity Model into Project Planning and Implementation

Utilize the insights gained from the initial benchmarking of the Digital Validation Maturity Model to guide the planning and development of a DVT. This includes considering different scenarios such as on-premise versus SaaS solutions, deployment strategies and their alignment with the overall success of DVT roll out. During the project charter and planning phase, factor maturity model considerations into decision-making processes. Evaluate different digital validation tools and methodologies to determine how they align with the current maturity levels and what is needed to advance to higher levels of maturity.

5. Measure, Analyze, and Continually Improve

Implement a structured approach to measuring the maturity of digital validation processes, analyzing results, and identifying areas for improvement. Regular assessments against the maturity model can highlight progress and pinpoint where further enhancements are needed. Establish a routine schedule for maturity assessments, using standardized tools and metrics to measure progress. Engage stakeholders in reviewing assessment outcomes and formulating action plans for addressing gaps. Document improvements and share successes to encourage ongoing engagement and support for digital validation maturity advancements.

Digital Validation Maturity Levels: Defining the Evolution from Paper-Based to Digital Systems

The journey from Level 1 to Level 5 in the context of digital validation maturity represents a comprehensive transformation of validation processes, evolving from traditional, paper-based methods to fully digital and optimized systems. This progression involves a shift in methodologies, technologies, and organizational culture to meet the increasing demands for efficiency, compliance, and adaptability in the face of rapid technological advancements and regulatory changes.

Level 1: Initial – Organizations at this stage rely heavily on paper-based documentation and traditional Waterfall methodologies. Processes are manual, with a heavy reliance on physical documents and multiple control points, which leads to inefficiencies and a higher probability of errors.

Level 2: Defined – At this stage, organizations begin to integrate digital tools with their primarily paper-based processes. While workflows are more controlled, the organization still relies significantly on manual testing and traditional methodologies.

Level 3: Maturing – At this transitional stage, digital execution becomes more prominent, and organizations start to adopt hybrid methodologies, blending traditional Waterfall with more iterative Agile practices. Digital execution becomes more prominent, and there is an increased focus on risk-based approaches. Automation is increasingly integrated with manual processes, marking a step towards optimization and more efficient workflows.

Level 4: Optimizing – Organizations at this stage have significantly adopted digital technologies, moving towards digital validation processes. The methodologies are highly flexible, with a mix of hybrid and Agile practices that allow quick adaptation to changes. The process improvements are substantial, aimed at increased efficiency and reduced reliance on physical documents.

Level 5: Transformed – At this stage, organizations have fully transformed their validation processes. Digital technologies, methodologies and automation are fully leveraged to create agile, responsive and automated validation systems. Automation and digitization are widespread, eliminating manual interventions and paper-based documentation.

The specific path and pace of this journey can vary greatly depending on the organization's specific validation portfolio and regulatory environment. Organizations focused solely on developing Software as a Medical Device (SaMD) or embedded software in medical devices may find themselves rapidly progressing through these levels, driven by the need for agility, continual improvement, and the high demand for repeated validation. In contrast, organizations primarily involved in drug product manufacturing may progress at a different pace, influenced by the distinct nature of their validation requirements, the stringent regulatory environments, and operational constraints.

The mix of validation types within an organization significantly influences its digital validation journey. In many cases, a hybrid model, incorporating elements from various levels, may be the best approach, combining their diverse needs, reflecting a bespoke/custom approach to achieving operational excellence and compliance.

The evolution from Level 1 to Level 5 is not only a technological transformation but also a cultural shift towards embracing digital innovation, risk management, and continual improvement as integral components of the validation process.

5.3 Planning and Developing Implementation Strategies

Several crucial elements should be carefully considered during the initial stage of implementation. Addressing these issues early on is essential for clearly defining the program scope and establishing effective implementation strategies. The following list, while not exhaustive, highlights key considerations that may vary depending on organizational requirements.

5.3.1 Rollout Scope Definition

The rollout scope must be clearly defined, whether the deployment is for a single site or a global implementation across multiple sites. Determine the rollout approach to be used, whether incremental (start with limited scope and expand gradually) or big bang (implementing across the entire organization at once). The chosen approach should align with organizational needs and objectives.

5.3.2 Outcome of Maturity Assessment

As emphasized in section 5.2.3, the insights derived from a maturity assessment serve as the cornerstone for a successful implementation strategy. A comprehensive understanding of the current organizational landscape, coupled with an awareness of the necessary steps to attain the envisioned state, lays the groundwork for crafting effective implementation plans.

5.3.3 On-Premise versus SaaS Solutions

A decision must be made regarding the adoption of an on-premise solution or SaaS model. This decision impacts the scope, implementation plan, and required resources for both implementation and ongoing support.

5.3.4 Scope of DVT Usage

The scope of the DVTs usage should be clearly defined (refer to Section 1.3), as this significantly influences the final project scope, plan, timeline, and resource requirements. Organizations may choose to utilize the DVT for specific purposes that support but are not part of the qualification and validation processes such as the overall QMS and Good Engineering Practices (GEP).

A clear understanding of intended use, and subsequent quality risk, and which specific regulatory requirements apply, is essential. This should be the basis of the lifecycle approach, the validation strategy, and defining which operational controls are required. The approach described in *ISPE GAMP® 5 (Second Edition)* [5] should be applied (see Section 5.6).

5.3.5 Integration Considerations

Determine whether the system will function as a stand-alone entity or if integration with existing software applications is necessary. Assess potential connections with systems such as Asset Management, quality management, financial, and project management.

5.3.6 Technology and Accessibility Considerations

- Identify all areas in which the DVT will be used
- Determine connection requirements for the DVT to be implemented
- Ensure that there is sufficient WIFI or hard-line access available in all areas to support personnel access
- Ensure that hardware appropriate to the areas is identified and obtained
- Prepare contingency plans for hardware failure

5.3.7 Sustaining Usage of Digital Validation Tool Post Implementation

It is important to consider how the organization will maintain the level of digital mindset and the use of the digital validation tool, in the post implementation phase. Having a site or organization champion is a valuable strategy to sustain the use of the digital validation tool. Their enthusiasm, expertise, and dedication can help foster a supportive environment where the tool becomes an integral part of daily operations.

- **Advocacy and Promotion:** the site or organization champion serves as the primary advocate for the DVT within the site. They continue to promote the benefits of using the tool.
- **Continual Support by organization leadership and stakeholders:** champions need the continual support of the stakeholders. For champions to be effective it is important for the role to be reinforced on a regular basis by the stakeholders.
- **Training and Support:** the champions also provide ongoing support, training, and assistance to users who have questions or encounter difficulties. By serving as a resource, they help ensure that everyone feels comfortable and confident using the tool.
- **Feedback and Improvement:** the site or organization champion gathers feedback from users about their experiences with the tool. This feedback is relayed to the project team or vendor to identify areas for improvement. By advocating for necessary enhancements or modifications, the champion helps ensure that the tool continues to meet the evolving needs of the site.

5.4 Stakeholder Engagement

5.4.1 Stakeholders

Thoroughly identifying key stakeholders is essential for the successful implementation of the DVT. This includes, but is not limited to, individuals from IT, quality assurance, production, regulatory affairs, and other departments directly impacted by or contributing to the validation processes. In addition, there may be external stakeholders such as end users, regulators, and clients in a CDMO setting. Identify site or organization champions who will help drive the digital transition and maintain the digital state post implementation.

5.4.2 Cross-Functional Teams

Consider establishing cross-functional teams comprising members from different departments. These teams can facilitate holistic discussions, enhance problem-solving capabilities, and ensure a well-rounded representation of perspectives throughout the implementation process.

5.4.3 Roles and Responsibilities

Clearly define the roles and responsibilities of each stakeholder within the context of the digital validation project. This promotes a clear understanding of expectations and accountability, fostering a collaborative and efficient working environment.

1. Project or Executive Sponsor (s):
 - Responsible for providing overall direction and support for the implementation of the project
 - Ensure alignment of the project with organizational goals and objectives
 - Allocate the necessary resources and support to the project team
 - Act as a liaison between the project team and senior management
2. Project Manager:
 - Lead the implementation of the project and oversee all aspects of the project
 - Develop and manage the project plan, timeline, and budget
 - Coordinate with stakeholders to ensure their requirements are met
 - Manage risks and issues that arise during the implementation process
 - Communicate progress updates to stakeholders and resolve any conflicts
3. IT:
 - Provide technical expertise and support for the implementation of the DVT
 - Install and configure the platform required for the tool
 - Integrate the tool with existing systems and infrastructure as necessary
 - Ensure data security and compliance with IT policies and regulations
4. Quality Assurance:
 - Ensure that the DVT meets regulatory requirements and industry standards
 - Collaborate with the project team to review and approve validation documents and other documents, ensuring alignment with quality requirements
5. Training and Documentation Team:
 - Develop training materials and documentation for the DVT
 - Develop training strategies and matrices
 - Conduct training sessions for end users
 - Update documentation as needed to reflect changes or enhancements to the tool

6. End Users:

- Provide input on their requirements for the DVT
- Participate in User Acceptance Testing (UAT) to ensure the DVT meets their needs
- Provide feedback during the implementation process to help refine the tool
- Receive training on how to use the tool effectively once it is implemented

7. Vendor/Supplier:

- Provide the DVT for implementation
- Offer technical support and assistance during the implementation process
- Supply validation documents and training materials
- Collaborate with the project team to meet specific requirements
- Provide updates and patches to the tool to address any issues or improve functionality

8. The **Process Owner** plays a critical role in ensuring the DVT aligns with business needs, compliance requirements, and process efficiencies. Refer back to section 2.4 for detailed roles of Process Owner.

5.5 Communication Channels

Establish robust communication channels to facilitate seamless information flow among internal and external stakeholders. Regular meetings, project updates, and feedback sessions should be structured to keep all stakeholders informed and engaged, fostering a sense of collective ownership. Use of asynchronous communication tools can enable extended team members and stakeholders to remain engaged and informed while streamlining meeting schedules and attendance.

5.6 Project Execution

5.6.1 Project Management and Organizational Change Management

This section underscores the pivotal roles of project management and organizational change management in the successful introduction and integration of a DVT. Project management serves as the backbone during implementation, focusing on logistical and operational aspects, ensuring technical adherence to specifications, timelines, and budgets. Conversely, organizational change management emphasizes the human element, managing cultural shifts, guiding staff through transition, and ensuring seamless integration. Effective leadership and stakeholder involvement, as discussed in Section 5.4, are critical for a successful initial phase, laying the groundwork for long-term success. Ongoing communication at various levels and the identification of local champions during implementation contribute to sustained program success, preventing a reversion to old practices.

5.6.2 Implementation

Once the project planning is complete and strategies are aligned within the organization, the project implementation phase can begin. The following are key considerations during this phase:

- **Business Process Optimization:** Review and enact improvements to the existing business processes to ensure compliance and minimize inefficiencies. This will enable the successful digitalization of the processes and the effective use of the resulting system.
- **Business Process Integration:** Align the DVT with existing business processes, ensuring that workflows are seamlessly integrated to optimize operational efficiency.
- **Documentation and Record-Keeping:** Establish robust documentation practices for the DVT. Documentation should be streamlined and aligned with risk-based approach, focusing on quality critical elements. While the records are primarily retained for internal quality assurance and process improvement, they also play an important role in demonstrating compliance with regulatory requirements, should an audit or inspection occur.
- **Data Security and Privacy Measures:** Ensure that effective data security measures are in place. Implement privacy safeguards to protect sensitive information in accordance with data protection regulations and organizational requirements.
- **User Training and Competency Building:** Develop user training programs to ensure employees are proficient in using the DVT. Assess and build user competency on a regular basis to enhance overall system utilization.
- **Change Management and Communication:** Introduce change management strategies to address organizational and cultural shifts. Create and implement communication plans to keep stakeholders informed throughout the implementation process.
- **Risk Management and Issue Resolution:** Identify and assess risks associated with the implementation. Develop risk mitigation strategies and promptly resolve any issues that arise during deployment.
- **Stakeholder Engagement:** As stated in Section 5.4, engage stakeholders at various levels to ensure alignment with project objectives. Encourage collaboration and open communication to address concerns and gain support.

5.7 Digital Validation Tool Lifecycle and Fitness for Intended Use

Each regulated company's intended use will vary, and there are numerous DVTs available, each with a unique set of capabilities, limitations, and degree of customization and/or configuration. Therefore, it is counterproductive to define a single, standardized method for implementing and validating a DVT. It is recommended that organizations adopt an incremental approach to implementation, coupled with a lifecycle strategy that leverages supplier activities and deliverables where possible.

Compliance with regulatory requirements and fitness for intended use can be achieved by adopting a lifecycle approach in alignment with good practices as defined in *ISPE GAMP® 5 (Second Edition)* [5]. This approach entails defining and performing activities in a systematic way from conception, through understanding the requirements, development, release, and operational use, to system retirement.

Verification throughout the lifecycle ensures that fitness for intended use is achieved, and that quality risks are managed to an acceptable level through appropriate controls.

The lifecycle for any system consists of four major phases: Concept, Project, Operation, and Retirement, as illustrated in Figure 5.1. For further details, refer to *ISPE GAMP® 5 (Second Edition)* [5], Chapter 4: Life Cycle.

During the concept phase, the regulated company should consider opportunities to automate one or more validation or qualification processes based upon business need and benefits. An initial risk assessment or GxP assessment should be conducted.

The project phase involves planning, supplier assessment and selection, specification development at various levels, configuration (or coding for custom applications), and verification leading to acceptance and release for operation. Risk management is applied to identify risks and mitigate them to an acceptable level.

Planning should cover all required activities, responsibilities, procedures, and timelines. For further details on computerized system validation planning, refer to Appendix M1 of *ISPE GAMP® 5 (Second Edition)* [5].

Lifecycle activities should be scaled according to:

- System impact on patient safety, product quality, and data integrity (risk assessment)
- System complexity and novelty (architecture and nature of system components, including maturity and level of configuration or customization)
- Outcome of supplier assessment (supplier capability)

Specifications should be comprehensive enough to support subsequent activities, including risk assessment, further specification and system development, verification (as appropriate), and system maintenance and updates. Verification activities, such as testing, confirm that specifications have been met. These activities may involve multiple review stages and testing depending on the type of system, development method applied, and intended use.

The system should be formally accepted for use within the operating environment and released into that environment through a controlled and documented process. Acceptance and release of the system for use in GxP regulated activities should follow a defined process and involve oversight and input from the process owner, system owner, and the appropriate quality function as necessary and applicable.

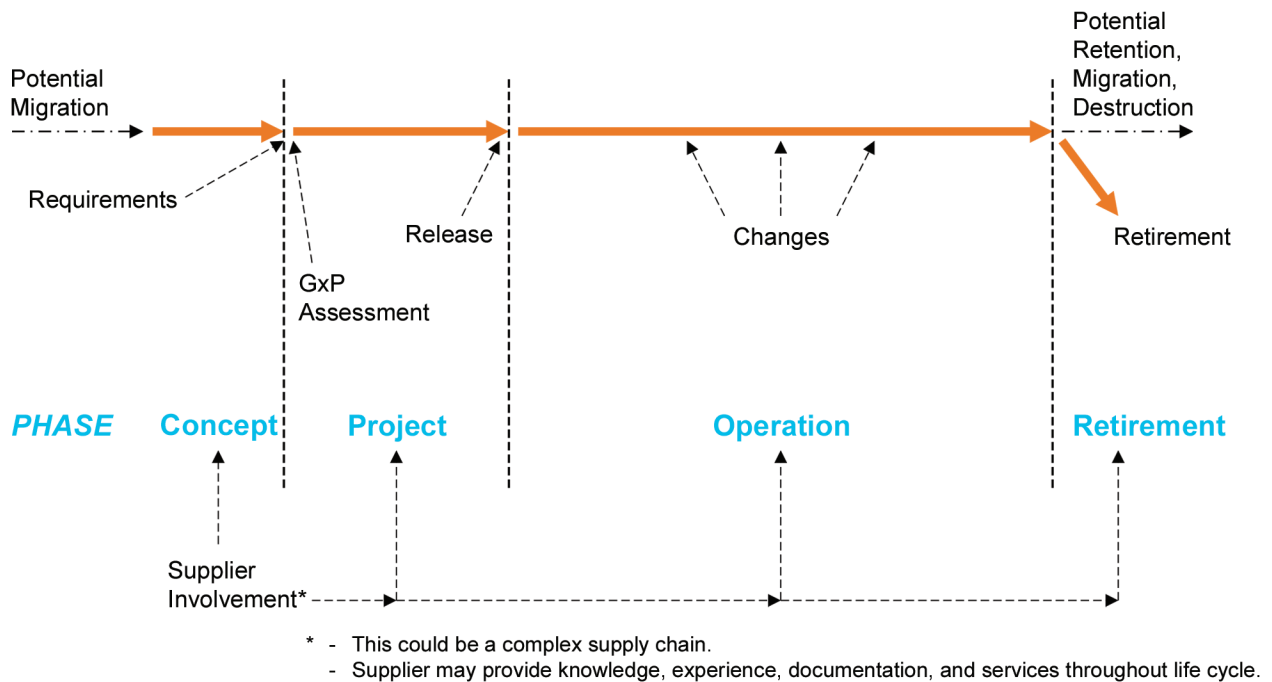
A computerized system validation report should be generated, summarizing the activities performed. For further information refer to Appendix M7 of *ISPE GAMP® 5 (Second Edition)* [5].

System operation is managed using defined, up-to-date, operational processes, procedures and executed by personnel who have the appropriate training, education, and experience. Maintaining control and managing change are important activities during this phase.

The final phase is the retirement of the system. It involves making decisions about data retention, migration, or destruction, and the management of these processes.

Suppliers of products and services should be involved as appropriate throughout the lifecycle. Many of the activities described can be delegated to suppliers, provided that the supplier assessment and control measures are satisfactory.

Figure 5.1: Life Cycle Phases [5]



The *GAMP* specification and verification approach is not inherently linear; it also supports iterative and incremental (Agile) approaches for product development, custom application development, and incremental product configuration. For further details see *ISPE GAMP® 5 (Second Edition)* [5], Appendix D8 – Agile Software Development.

Most DVTs are SaaS based, and particular attention should be given during the supplier assessment to ensure that cybersecurity risks have been adequately mitigated, and that the supplier has appropriate procedures in place for disaster recovery and data backup commensurate with the intended use and level of risk. Refer to the *ISPE GAMP® Good Practice Guide: Enabling Innovation* [10], Chapter 4.

5.8 Deployment and Integration

5.8.1 Communication

Execute the communication plan to ensure all stakeholders are informed about the deployment schedule, expectations, and support mechanisms.

5.8.2 User Support

- Conduct training sessions to familiarize end users with the new system and address any concerns
- Provide dedicated support during the rollout phase to assist users in adapting to the new system
- Provide ongoing support for users after rollout to maximize adoption and effective use of the system

5.8.3 Feedback Mechanism

- Implement a feedback mechanism to collect insights from end users during and after the rollout
- Use feedback to identify areas for improvement, address user concerns, and enhance user experience

5.8.4 Integration with Existing Systems

- Conduct a thorough assessment to ensure compatibility between the DVT and existing technologies
- Identify potential integration points and dependencies with other systems
- Develop interfaces or connectors to seamlessly integrate the DVT with existing software applications
- Ensure data flow and consistency across interconnected systems
- Execute migration processes with caution, ensuring data accuracy and integrity
- Implement rigorous testing of integration points to identify and resolve any issues
- Conduct both automated and manual testing to validate data exchange, system synchronization, and overall interoperability

5.8.5 Retirement, Archiving, and Migration

During system retirement, the system is removed from active operations, normal operational users are deactivated, and interfaces disabled. No additional data is entered into the system after this point. Special access may be retained for purposes such as data reporting, results analysis, and support.

Decommissioning refers to the controlled shutdown of a retired system. Data and documentation must not be disposed of until they have met the record retention period specified in the organization's record retention policy. The process to be followed for system retirement should be documented in a system retirement plan.

The plan should specify which data will be migrated, archived, or destroyed, as well as the associated approval process. The approach to data migration and archiving should be determined based on the anticipated frequency of access, the need for reprocessability, and the record risk level and media robustness. Controls should be established to ensure that records and data remain secure, complete, and accurate, and that signature/record linking is preserved, where applicable.

See *ISPE GAMP® 5 (Second Edition)* [5], Appendix M10 – System Retirement, Appendix O13 – Archiving and Retrieval, and Appendix D7 – Data Migration for further details.

6 Digital Validation Tool Governance

6.1 Introduction

When implementing a digital solution, key value drivers identified are often focused on improving efficiency and enabling users to complete activities in a compliant manner. Good governance should add value while not introducing additional levels of bureaucracy to the digitized process.

The governance of GxP regulated computerized systems is outlined in detail in *ISPE GAMP® 5 (Second Edition)* [5], Section 6.1, Governance for Achieving Compliance.

When deploying a DVT, it is important to agree and clearly communicate acceptable and unacceptable practices and then ensure adherence to these practices. A visible, robust governance process accelerates deployment, simplifies operations and reduces variability in the digitized process. A consistent process, paired with standardized data, enables the company to maximize the benefits of shared information and provide a solid framework for future process and technology advancements.

This chapter addresses the necessary level of governance to optimize the benefits of using a DVT:

- Governance Team
- Governance Approaches and Considerations
- Managing Change
- Procedures
- Training
- Performance Monitoring

Consistent processes and terminology are essential to support knowledge management and leverage reporting data to effectively share information across the business. These should be established as part of system implementation (refer to Chapter 5), with governance processes in place to ensure compliance.

Changes to the scope of the processes being digitized, as well as adjustment to the user base, impact funding requirements. Therefore, these changes need governance agreement and support at a senior-level.

6.2 Governance Team

Formal governance structures must be in place to provide clear leadership and accountability for change management, as well as tracking and reporting benefits. While these governance structures should be formal, they may often be supported by less formal meetings or communications to strengthen and share best practices.

The structure of the governance team varies depending on the company, as well as the scale and scope of the deployment. Below is an example of roles and responsibilities:

Sponsors and Stakeholders: to ensure rapid delivery, senior stakeholder engagement is essential. Typically, for a validation process this includes the Process Owner, System Owner, IT, Quality, Manufacturing Technology, and Engineering teams. Sponsors need to be clear on the expected scope and timelines for deployment and should be prepared to intervene to ensure the objectives are met.

Process Owner: this role is important and required for the effective implementation and ongoing ownership of the tool. This individual is key to ensuring the tool is utilized in a way that supports organizational goals and compliance.

Project Team: this team is responsible for delivering the chosen solution along with the necessary supporting documentation and training materials. This team typically includes the IT function in addition to those directly supporting deployment. It is important that those supporting the deployment understand the process being digitized. After the implementation phase, part of the project team should remain in place to manage ongoing changes and improvements.

Supplier: A supplier representative supports the necessary supplier assessments and audits, provide feedback to supplier on issues encountered, and address product development requests. Suppliers also provide visibility into their product roadmap. Where a SaaS deployment model is used, suppliers ensure availability and security of the data on the system.

Site Deployment Champions (Super Users): Experience has shown that the site champions, who are often the early adopters, play a crucial role in successful early implementation. At sites with established legacy processes, a single point of contact is important for addressing changes to existing workflows.

These Site Deployment Champions should have a clearly defined role and be measured against achieving deployment objectives.

Users: A positive user experience is vital for adoption and a useful source of feedback. Establishing a Community of Practice (COP) allows the project team to interact with users, communicate changes, and share learnings. Understanding the user profile, whether they are specialists, production operatives, or equipment suppliers will help provide targeted governance.

Encouraging early adopters to share the benefits they have experienced is often more impactful than presentations from a central Project Team, as peer testimonials are often more persuasive.

6.3 Governance Process

6.3.1 Governance Models (*Global versus Local*)

It is important to adopt the governance model that best aligns with the size and deployment objectives of the company. Consider whether the objective is to drive a standardized process across multiple sites or to provide a flexible tool that allows sites to adapt and develop their own processes.

If the intent is to change process ways of working at the same time as deploying a digital solution, it is important to communicate this clearly to sponsors and stakeholders, as this change requires additional governance and may complicate or slow the deployment process.

Two governance models are described in Table 6.1. A hybrid approach may also be considered, where core documents are controlled centrally with the option of other site-specific documents configured locally or where there exists differing regulatory entities and requirements. The governance model should be agreed upon as part of the implementation process as it impacts the governance structure during routine operations.

Table 6.1: Example Governance Models

Model	Approach	Positives	Negatives
1	Global control over templates, a small group maintains the content.	Consistent ways of working delivered by a small team.	Deployment progress may be slower as new ways of working need to be adopted if not already consistent across all sites. Lost opportunities for inclusion of “local” forms that could be included in the scope.
2	Local sites or departments have the ability to create local templates.	Fast response to changes; documents can include site-specific verbiage. Resources to make updates are held at sites. Improved ownership and take up.	Database can get very complicated, loss of control and standardization. A higher level of training is required for site users. With more users having higher access rights within the system, the change management process becomes more onerous.

Figure 6.1 provides an example of a governance structure. This example shows a central product team responsible for the product and process owners that manages the defined configured processes.

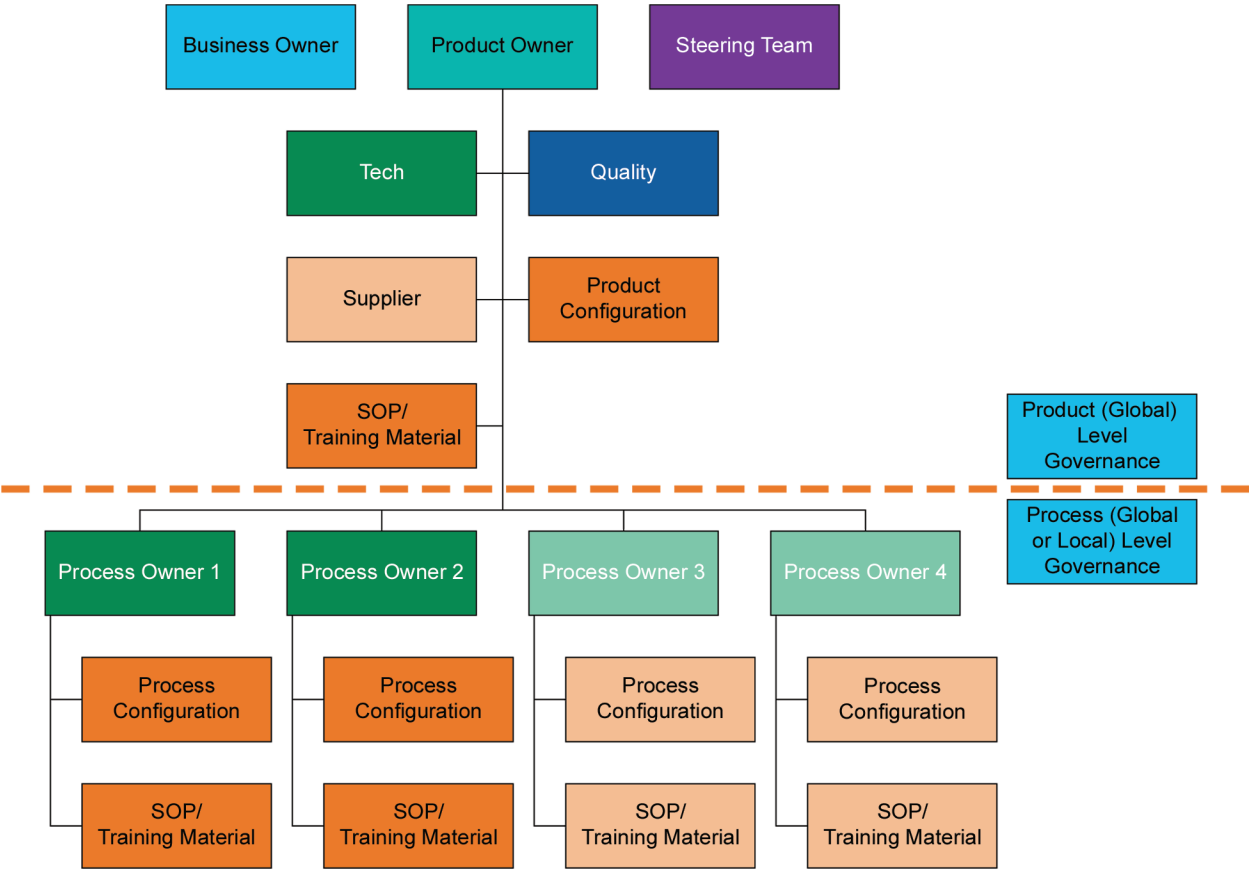
In this model the Product Owner is responsible for product configuration, procedures, and supporting material. Each Process Owner is responsible for the process-specific configuration, procedures, and supporting materials.

The Process Owners may be organized by:

- Global Process (Equipment Validation, Cleaning)
- Local (site-specific forms)
- Product (Products or dose forms)
- Customers

The Processes could be a defined global validation process or a site-specific process.

Figure 6.1: Example Governance Structure for a DVT



6.3.2 Governance Board

There should be a governance hierarchy, with meetings held to agree on changes and manage escalation items. Meetings may be held at the product (global) level or at the process level. Meeting frequency should be determined based on the maturity of the organization, the deployment, and scope of the program. Table 6.2 provides examples of structure and schedules.

Table 6.2: Example Governance Meetings

Meeting Type	Level	Frequency	Attendees	Purpose
Steering	Product	6 Months	Sponsors and Stakeholders	Feedback deployment metrics and escalation items
Supplier	Product	3 Months	Supplier and Project Team	Discuss roadmap, issues and feature requests
Governance	Product/ Process	Monthly	Project Team	Agree change requests
User Community of Practice Meeting	Process	Monthly	Project Team and Super Users	Share good practice and information, request changes

6.3.3 Governance Plan

The governance model to be followed should be defined in a governance plan, which should be subject to routine revision:

- Verify key contacts
- Review monitoring metrics
- Document any identified compliance issues as risks or exceptions
- Review any existing exceptions or risks related to the third-party services and update remediation plans and actions as necessary
- Verify that company information is managed in accordance with record retention requirements
- Other, including any key risk indicators that have been defined
- Complete periodic validation review/periodic evaluation for consistency with the regulations, if applicable
- Ensure Business Continuity Plan (BCP) is in place and current – refer to *ISPE GAMP® 5 (Second Edition)* [5], Appendix O9 Backup and Restore and Appendix O10 Business Continuity Management

Example monitoring metrics:

- Downtime (system unavailability)*
- Incidents volume
- Requests volume
- Review of open problems
- Service Level Agreement (SLA) compliance
- Quality agreement monitoring

***Note:** System unavailability has a high impact; therefore, it should be prioritized as a key performance metric to monitor and track.

6.3.4 Governance Considerations

6.3.4.1 Access Controls

As outlined in the *ISPE GAMP® Guide: Records and Data Integrity*, Section 2.2 Data Integrity Requirements [15], regulated companies should apply access controls to ensure that individuals only have access to functionalities relevant to their job roles, and that actions are attributable to a specific individual. These controls should prevent the unauthorized deletion or modification of regulated data and records, whether within or outside the software application (e.g., limiting user rights).

“Regulated companies should be able to demonstrate the access levels granted to individual staff members and ensure that historical information regarding user access level is available. User access arrangements should prevent (or if necessary, log) unauthorized data amendments.” (RDI)

An access management plan should document the access process, clearly communicating the approach to both internal stakeholders and auditors. This plan should balance the requirements to meet the compliance (and business) needs of robust access control while allowing users to be onboarded quickly. For an ad hoc/temporary users (e.g., from operations) who need to document a test result (a one-time task), a quick and easy onboarding process is crucial; therefore, make the access process as simple and quick as possible.

When determining the access management process, also consider if access will be allowed to company personnel only or external contractors and consider any confidentiality requirements. Consider if the tool and structure selected supports the required level of confidentiality (access limited to certain users or groups), or knowledge sharing requirements (where access may be required internally and externally).

User access reviews should be completed in line with regulated company policy, e.g., every 12 months, although this frequency may vary depending on data criticality. Roles with higher level functionality, such as System Administrators may be reviewed and challenged more frequently.

For further information, see also *ISPE GAMP® 5 (Second Edition)* [5], Section 4.3.7 Security and System Administration.

6.3.4.2 Data Lifecycle Management

The *ISPE GAMP® Guide: Records and Data Integrity* [11], Chapter 4, Data Life Cycle, outlines a generic data lifecycle model suitable for managing all types of data and records. It also defines the activities and requirements for each phase in the data lifecycle and variations within these phases for different record types. Clear data ownership is required and will form part of the company's wider data governance framework.

Considerations specific to DVT include:

- If the DVT is to act as a repository of information, it is important to designate data owners and establish an archival approach for storing or deleting data that is outside of its retention period
- A data transition plan describes how the transfer of data is managed:
- Data from legacy systems: consider how legacy validation data (e.g., RS or DS) will be incorporated. Will this data remain in its legacy format and location, or will it be transitioned to the DVT? Does the selected DVT support this process?
- For a service provider, how will data be transferred to the end user in an acceptable format?
- In the event of the sale of the regulated company or the service provider, how will data be transferred from the DVT to the new owner?
- What is the process for sharing data with other company systems?

6.3.4.3 Internal Auditing

Internal business management auditing supports the governance process in ensuring compliance with procedures and data standards. This monitoring is made easier by the digitization of the process and accessibility of reported data.

Audits can be used to confirm:

- Adherence to defined processes and templates without unnecessary local customization
- Compliance with established numbering and naming conventions

- Achievement of expected usage levels as identified in the business case
- Identification of any usage outside of intended and configured processes
- Completion of access reviews
- Identification of good practices that should be shared across the organization

Internal audit teams play a key role in ensuring the agreed templates and standards are consistently followed.

6.3.4.4 Communication

The Governance Team should ensure the regular communication of deployment successes and benefits realization to the wider business. This supports the ongoing funding and helps identify additional use cases for the DVT. Recognizing personnel who supported the change process is also important for maintaining momentum and engagement.

6.3.4.5 Managing Change

In this section recommendations for the management of changes to a DVT using a risk-based process are proposed. This includes the technical and procedural controls that could help mitigate the risks associated with these changes.

Examples of types of changes considered include:

- Change of use (e.g., reconfiguring the system for a new process)
- Software updates or patches
- Product configuration changes – related to technical controls within the system (e.g., audit trail frequency or user roles)
- Process configuration changes (modifications to templates and process configuration within the system)

As described in *ISPE GAMP® 5 (Second Edition)* [5], Section 4.3.4 Change Management, changes that are proposed during the operational phase of a computerized system should be subject to a formal change control process. This process ensures that proposed changes are appropriately reviewed to assess the impact and risk of implementing the change. In some cases, regression analysis and regression testing may be required. *“The process should ensure that changes are suitably evaluated, authorized, documented, tested, and approved before implementation,”* with the process formally closed once the change is made.

The rigor of this approach, including the extent of documentation and verification, should be scaled based on the impact of the change on the quality of the supported process. For instance, tools supporting process validation have a higher impact on patient safety and product quality than those supporting C&Q. Similarly, tools supporting the medicinal product lifecycle are of greater significance than those supporting system or equipment lifecycles.

The process should allow flexibility in scaling the rigor of documentation and verification, based on the nature, risk, and complexity of the change, applying critical thinking at each stage.

For further guidance, see *ISPE GAMP® 5 (Second Edition)* [5], Appendix O6 Operational Change and Configuration Management, and for patch and update management refer to *ISPE GAMP® 5 (Second Edition)* [5], Appendix S4 Patch and Update Management.

6.3.4.6 Types of Change

Process Change or Change of Use

A process change refers to the introduction of a new validation process that requires the DVT to be used for a new process that it was not originally configured or validated for. For example, a DVT initially validated for risk-based C&Q might be extended to support process validation. This change can significantly affect patient safety and product quality as the validation scope shifts from lower risk record/data types (risk-based C&Q records) to higher risk data types (such as intellectual property or restricted/confidential information). Also, from a GMP data classification perspective, the process validation records could be considered a higher risk level, as these records could be used as part of regulatory submissions and are readily audited.

The change in the data type could lead to additional validation of the DVT to ensure the following:

- The data retention/disaster recovery policies are met for the new data type
- A formal risk assessment of the use of the DVT, and determine further validation deliverables
- More frequent review of access control

Due to the higher risk associated with this type of change, it is recommended that such changes be rigorously assessed through the QA change control process to determine the impact.

This assessment should include the business owner and be managed at a global level to ensure the processes/system changes are implemented and communicated to all users.

The change management process for process configuration changes is simpler if a global model (Model 1 – see Table 6.1: Example Governance Models) is adopted. However, if a site-based model (Model 2 – see Table 6.1: Example Governance Models) is used, strong governance is required, as many more users typically have access rights to make changes.

Configuration Changes

Configuration changes can be categorized as:

- **Product Configuration Changes:** these include technical controls within the system (e.g., enabling user access roles, modifying audit trail settings, adjusting acceptable file types)
- **Work Process Configuration Changes:** changes to the DVT involved in the set up of validation processes/templates/approval workflows, e.g., templates or approval workflow modification

These two types of change have different risk levels and the approaches to managing these changes.

Changes to the product configuration should be managed through the QA/technical change control process as they are directly associated with access and security controls.

Work process configuration changes may be controlled procedurally relying on system technical controls and approvals. These types of changes can be applied either globally or locally within the system, and the change control process followed may depend on which level is affected.

Table 6.3: Example Approaches for Change Control for DVT Configuration Changes

Work Process Configuration Changes	Product or Process Change	Risk and Rationale	Technical Controls	Procedural Controls
User Access Level Change	Product	Untrained individuals can make setup /configuration modifications	Audit trail Access control (only administrators can make this change)	Access request process Periodic review of access
New process	Product	New process use case may require additional product controls	Access control (only administrators can make this change)	Review of documents upon approval
Modified process	Process	Modified process not configured correctly leading to wrong documents used, less efficient data sharing	Access control (only administrators can make this change)	Review of documents upon approval SOP for change control of DVT
New custom report (for GxP purposes, i.e., RTM)	Product	Untrained individuals can make setup/configuration modifications	Access control (only administrators can make this change)	SOP for change control of DVT
Document template changes	Process	No unauthorized change can happen due to approval of changes required in DVT	Audit trail Approval of template in DVT Access control (only certain people can make updates)	SOP for change control of DVT RACI for approval of documents states to check content to SOP
Document approval workflows changes	Process	Modifications to approval templates is only assigned to trained individuals and approvals are checked on each document to SOP on approval	Audit trail Access control (only certain individuals have permissions)	SOP for change control of DVT Approval of documents confirms to SOP requirements, and this is checked on approval by QA
Folder/filing structure changes	N/A	Data is not changed; some data sharing or reporting could be affected leading to less efficiency using tool	Audit trail Access control (only certain individuals have permissions)	SOP for use of DVT

6.3.5 Standard Operating Procedures (SOPs)

A key component to this governance model is the development of SOPs. These procedures establish standardized processes for using the DVT, ensuring consistency in operations across different sites and user groups. This consistency is important for maintaining compliance and enhancing efficiency. While the implementation team will have developed the initial SOPs (see Chapter 5), these will need to be updated by the Governance Team as system changes are made and system knowledge increases.

Impact of Changes to Existing Procedures and Ways of Working

In addition to the creation of new procedures for the DVT, it is important to assess whether changes are required to the existing legacy validation controls developed for paper-based processes and determine if they are still appropriate, for example:

- With paper-based test protocols, paper attachments are numbered, initialled, and dated and with a reference to a test case or protocol. In a DVT system, where exhibits are digitally controlled to prevent tampering, this manual annotation may no longer add value as a control and may simply extend the testing time.
- Some companies developed working practices where the Quality approver was required to be the final approver on a document. In a digital system with built in technical controls to lock content from changes after approval, this step may no longer be required and could delay the final acceptance and release of the system.

The following are examples of items that benefit from a global or central procedural approach. These aspects should be included in global SOPs:

- Structure, naming conventions, and numbering: establish clear standards for these elements where they are not automated
- Data management processes: define standardized data management protocols across the organization
- Reporting expectations and user confidentiality: establish procedures for handling reporting and ensuring user confidentiality

6.3.6 Training

The approach to training must accommodate a diverse user group while ensuring compliance with regulatory standards. It is important to have a training plan that is both robust and simple to execute. To ensure effective training, consider creating a set of global SOPs, creating role-based training for different user groups, setting up a training environment within the system, and adhering to best practices when establishing training.

- Role-Based Training for Different User Groups

Role-based training is a cornerstone of this model. Users of the DVT will vary in their roles and responsibilities, necessitating tailored training modules. For instance, those who are simply approving documents or recording test results require far less training than those developing test documents. Customizing training content to suit these varied system roles allows users to acquire the necessary skills and knowledge to effectively use the tool in a time efficient manner. Each training module should also provide all users with an overall understanding of the general workflow of data within the DVT.

- Training Environment Within the System

Providing a training environment that mirrors the production environment allows users to engage in hands-on practice without impacting the actual production environment. The training environment should be designed to simulate real world scenarios, enabling users the opportunity to gain practical experience in a controlled, risk-free setting. Time spent in the training environment before the “go live” phase helps alleviate concerns and reinforces other forms of training provided.

- **Good Practices in Establishing Training**

Training should be flexible enough to address the needs of the different users and roles within the system. The formal procedures established should contain the essential steps that each user must follow, while interactive and engaging training methods such as simulations and quizzes can enhance the learning experience. Ensuring the accessibility of training materials for users in their local language is highly beneficial, although it can place additional demands on the support team. When functionality is more “locked down” and aligned with specific user roles within the DVT, the simpler the training becomes.

- **Innovative and Informal Training**

In addition to formal training methods, incorporating informal training tools can significantly boost user competence and confidence. Creating a resource such as Wiki or FAQ section allows users to quickly access key information and resolve common issues. Developing short video tutorials or clips on specific topics (ideally provided by the DVT supplier) can serve as quick refreshers, aiding the retention of key information. These tools provide material that can be easily updated or enhanced without making changes to training plans or more formally controlled training material.

- **Competence Verification**

It is important to have a method in place to verify the training has been fully understood. For ad hoc users who only need to document test results, a brief onboarding training may be sufficient. However, for more involved users, such as those starting a large capital project, it may be beneficial to allocate time for them to develop a set of dummy deliverables under supervision in a training environment. This ensures that efficiency improvements are understood and applied.

Implementing a comprehensive governance model for training is essential to ensure the effective use of the DVT.

6.4 Performance Monitoring

6.4.1 Reporting

The system selected and the data structure created (e.g., process, numbering and naming conventions, entity structure, etc.) will impact the quality and ease of the reporting available.

Performance monitoring may be split into three categories:

- **Product (DVT) Focus:** Is the DVT being used as expected?
 - Number of active users/last access date
 - Number of approved documents per month
 - Number of templates used
 - Templates due for review
- **Process Efficiency:** What data is needed to understand and improve the efficiency of the digitized process?
 - Document development durations
 - Test durations for specific entity types
 - Document review and approval durations

- Number and type of test failures identified
- User Level: What data allows managers or users to perform their tasks more effectively?
 - Number of documents pending approval (by person, department, or system)
 - Resource loading
 - Task duration or percentage complete
 - Identification of long-term open or abandoned documents
 - Validation status of the entities
 - Documents next due for review or entities due for requalification

6.4.2 **Benefits Realization**

The governance team should track and report on the delivery of expected benefits outlined in the business case. This review is critical for demonstrating how the implementation of the DVT has benefitted the organization, and to determine where further work is needed to alleviate bottlenecks or maximize efficiencies. Benefits should be tracked in the early phases of deployment.

How to Measure Benefits without Baseline Metrics

If baseline metrics from the original paper-based process are unavailable, one approach is to use anecdotal comparisons from users following initial deployment. Early users can compare test and approval durations, data integrity improvements, ease of use, and any other metrics.

- **Track from the Start:** Use a simple form to capture feedback for each initial project.
- **Identify Savings:** Measure total duration savings as well as work hours consumed. Earlier completion dates are often provide the most significant benefit, as testing can be reviewed as it progresses and in parallel rather than sequentially, as in paper processes.
- **Quantify Simplification Benefits:** When measures are taken to simplify processes during digitalization, attempt to quantify the benefits associated with this simplification as all efficiencies may not be directly linked to the digitalized change.
- **Leverage New Data:** Once implementation is complete, use data such as the number of approved documents, as well as review and testing durations. This new data should be incorporated into routine system review processes and provided to the governance team as indicators of ongoing system health.
- **Continual Improvement:** Look for opportunities to use the increased availability of the data to further enhance identification of continual improvement initiatives and knowledge management.

Once sufficient data is available, communicate the results to stakeholders and the wider company.

6.4.3 **Benchmarking, Optimization, and Improvement**

Effective governance should incorporate both internal and external benchmarking to continually optimize and improve workflows.

The large amount of performance data now available allows for comparisons to be made across sites and teams following similar processes. This data can highlight best practices to share or identify bottlenecks that require investigation.

Data that was rarely captured in the past (e.g., test durations, approval times) can now be utilized as a source for future project schedules, to provide better planning assumptions.

Consider comparing the organization's performance with other similar companies to identify opportunities for mutual learning and improvement.

Consider the future advancements in technology when structuring and managing data so that its value can be maximized. This may be as simple as the reuse and refinement of digitized test scripts or as a reliable source of data for an AI tool.

6.5 Accurate and Usable Validation Records and Copies

Maintained validation records and data must preserve both content and meaning while meeting predicate rule requirements. These records may be created within the DVT or consist of accurate and usable copies of records and data created or captured elsewhere. They must be maintained in a manner that allows for the understanding, review, and retention of the information.

The DVT must support the management of all stages of the record and data lifecycle, from creation to destruction. It should be capable of handling many different types of data, including both static and dynamic data. For further reference, see *ISPE GAMP® RDI GPG: Data Integrity by Design*, Section 3, Data Retention Strategy [14], and *ISPE GAMP® RDI GPG: Data Integrity – Key Concepts*, Section 3 Data Life Cycle [12].

A sufficient level of control can be achieved by ensuring that the attachment in the digital validation system is scanned and aligned with ALCOA principles. When approvers sign off on the verification document, they are also responsible for confirming that the evidence is complete, accurate, and meets the acceptance criteria of the verification. This approach provides robust control, as the attachment is not editable once stored in the DVT without an audit trail, and this audit trail will be reviewed as part of the verification documentation approval.

The process of verification should, however, be risk-based, taking into account the criticality of the records, the maturity of the organization, and other technical and procedural controls in place.

Example:

For operational verification attachments from dynamic data like temperature trends from validated systems such as data historian or process control systems, which may include metadata (e.g., applied filters, scalability factors, and audit trails), the following practices are recommended:

1. Print the trend, to a PDF (if possible), and if required, annotate the data with reference to the validated data repository. Attach this as a true copy into the digital validation system under the associated verification test case.
2. Capture in the digital validation system the reference needed to re-create the data in the validated system (if not annotated). This ensures that Metadata can be reviewed in the validated system (e.g., an API based reference or link to the validated system).

For data originating from an unvalidated system, provisions should be made during the verification process to store and reference the data, allowing it to be re-created in the future.

7 Regulatory Considerations, Inspection Readiness, and Implementation

7.1 Standards and Guidelines

Digital validation operates within a complex regulatory environment, and GxP requirements as well as expectations must be addressed. Although digital validation is not specifically outlined in international regulations, several related requirements still apply. For instance, critical validation records must comply with EU documentation requirements [23] as well as equivalent US FDA GMP requirements [24], Subpart J Records and Reports. Additionally, ALCOA++ principles as described in Chapter 3 of this ISPE Guide and Chapter 4 of EU GMP, must be met. DVTs must be compliant, fit for intended use, and meet the requirements of EU GMP Annex 11 [7] and US 21 CFR Part 11 [1]. These requirements can be met by adopting a lifecycle and governance approach based on *ISPE GAMP® 5 (Second Edition)* [5] as described in Chapters 5 and Chapter 6.

The US FDA's Data Integrity and Compliance with Drug CGMP, Questions and Answers, Guidance for Industry [25] states that, *"CGMP regulations and guidance allow for flexible and risk-based strategies to prevent and detect data integrity issues. Firms should implement meaningful and effective strategies to manage their data integrity risks based on their process understanding and knowledge management of technologies and business models."* Such strategies form the foundation of the approach presented in this ISPE Guide.

As stated in ICH [26], new technologies are continually being developed and should be adopted when they are considered to offer additional assurance of quality or are otherwise justified. Similarly, the FDA notes that drug shortages and product recalls are often linked to *"compromised product or facility quality (e.g., the utilization of outdated technologies and equipment)."* The FDA aims to encourage and sustain manufacturing quality improvements through innovative approaches, which are critical to modernizing the pharmaceutical manufacturing landscape and improving quality [27].

Digital validation is an example of such an innovative and valuable approach. Carefully applying the guidance in this Guide, can help companies achieve compliant and fit-for-purpose DVTs. By staying ahead of regulatory trends and fostering collaboration with regulators, companies can ensure they remain compliant while fully leveraging the capabilities of DVTs.

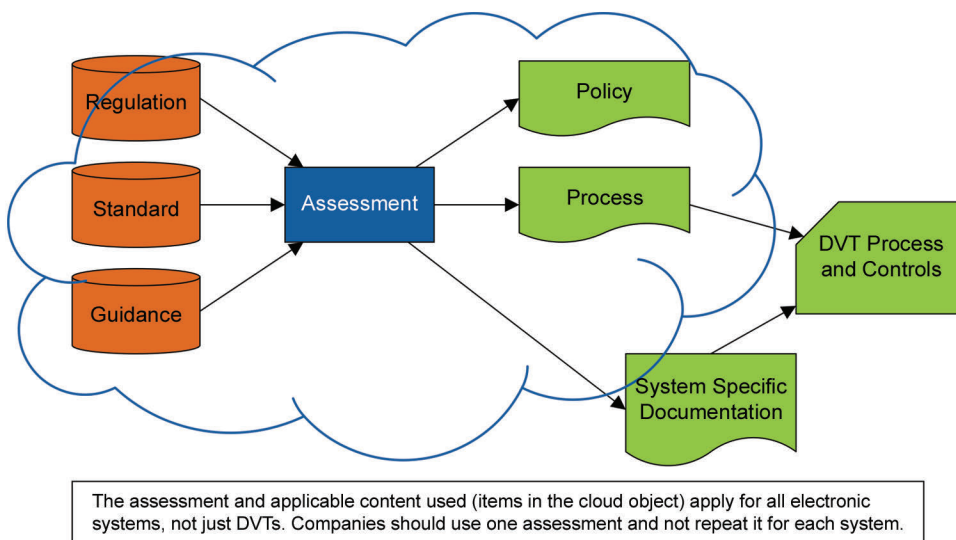
7.1.1 Regulatory and Standards Assessment

Within a complex operating system that can span multiple countries and regulatory jurisdictions, the industry needs a governance strategy that allows regulations, standards, and guidelines to be connected in a matrixed environment.

Understanding of the regulations and their necessary controls can be found in the company frameworks, processes, and the deliverables for those processes. A straightforward way to accomplish this is by performing an assessment of the applicable regulations and standards and mapping them to the procedural and system-specific documentation.

Performing an assessment can enable the company to manage all these overlapping regulations, standards, and applicable guidances in a way that allows for cross-linking to avoid duplication of processes for different areas.

Figure 7.1: Regulatory and Standards Assessment



7.2 Internal Audits and Regulatory Inspection Techniques and Challenges

7.2.1 Introduction

Inspectors need to trust that an organization is providing accurate information efficiently, without intentionally delaying the inspection process. If a DVT is poorly presented during an inspection, it can undermine the organization's credibility. In extreme cases, obstructing an inspector's access to information or presenting misleading information or data may be considered a criminal offence in some regions.

To mitigate these risks, it is essential for organizations to plan carefully how they will present validation data during an audit. A collaborative approach between the organization and the inspector can be very effective. For example, an SME could directly present data from the DVT system upon the inspector's request. This allows the organization to provide context and better address any potential deficiencies in the validation process.

Discussing the methods of accessing and presenting the information with inspectors beforehand or during their visit can help streamline the inspection process and prevent misunderstandings.

7.2.2 Evolution of Validation and Inspection Processes

Regulatory inspections have undergone significant evolution, particularly in the validation and inspection processes. These changes are driven by technological advancements, increased global collaboration, and the need for more robust compliance mechanisms.

Historically, regulatory inspections focused on end-product testing and adherence to basic manufacturing practices. However, as the life sciences industry has grown in complexity, the emphasis has shifted towards a more holistic approach considering the entire production lifecycle, from raw material sourcing to manufacturing and final distribution.

Today, validation processes incorporate advanced technologies such as real-time data monitoring and analytics, which allow for continuous oversight of production processes. These technologies help identify deviations from standard procedures promptly, allowing issues to be addressed before they affect product quality or safety. Additionally, risk management principles now play a central role in the validation process, guiding the allocation of resources and controls to areas with the highest potential impact.

The global nature of the pharmaceutical industry has also influenced the evolution of inspection processes. International regulatory bodies such as ICH [28] have worked towards harmonizing standards. This global approach helps streamline processes for companies operating in multiple countries, ensuring consistent quality and compliance.

Despite advancements, challenges remain, particularly in adapting to rapidly changing technologies and managing increased data volumes. Future developments are likely to focus on integrating artificial intelligence (AI) and machine learning to further enhance the efficiency and effectiveness of regulatory inspections and validations. This evolution not only supports compliance but also fosters innovation within the life sciences industry.

Another significant change in recent years is the increased use of remote inspections. The same activities, processes, and systems are examined during a remote inspection as those in on-site inspections, although the processes, tools, and techniques to support remote inspections differ. For example, presenting validation records and data that is maintained in paper-based systems can be challenging during a remote inspection, requiring either physical or electronic copies of the paper records to be provided. On the other hand, records and data maintained in a DVT can be presented more easily during remote inspections by leveraging teleconferencing technology, or by granting regulators direct access to the DVT.

7.2.3 Remote/In-Person Inspections

Whether the inspection is remote or in-person, various strategies can be employed to present both the DVT and the records and data contained within it. In general, validation records and data should be presented from the system of record. The organization must define where the original documentation is stored (paper format, DVT, DMS, or other location/system)

- If the system of record for validation activities has been defined as physical documentation (paper), then the audit and inspection strategy should plan to present the physical documentation and explain the control and retention processes related to those documents.
- If validation activities are executed in the DVT but the system of record has been defined as the DMS (or other mechanism), then the audit and inspection strategy should plan to present information from the DMS. In this scenario, the controls and processes for transferring data and maintaining retention across both systems will be important.
- If the DVT has been defined as the system of record for validation activities, then the audit and inspection strategy should plan to present information directly from the DVT.

There are several strategies for the presentation of records and data from the DVT as the system of record. Table 7.1 outlines the major advantages and disadvantages of each.

Table 7.1: Major advantages and disadvantages of DVT as the system of record

Pros	Cons
Strategy 1 – Provide open system access (inspector is provided a full user account within the DVT)	
• Presents records and data directly within the system of record	• Time intensive – requires inspector to complete training per current DVT access requirements • Requires absolute confidence in DVT design/configuration, content, and ability of SME to speak to any record accessed by the inspector • Provides the lowest level of control by the organization during an inspection
Strategy 2 – Provide limited system access with clearly defined boundaries/information map (inspector is provided a limited user account within the DVT)	
• Presents records and data directly within the system of record • Inspector can navigate and fully experience the DVT design and use • Demonstrates clear confidence in data integrity and DVT functionality, may lead to deeper discussions regarding the validation activity rather than the DVT function • Provides remote access capability for continued review after-hours • Demonstrates transparency and builds trust with the inspector	• Time intensive – requires inspector to complete training per current DVT access requirements • Time intensive – requires organization prework to create correct/relevant access and define (configuration, account creation) • System functionality dependent – requires granular access and data segmentation controls
Strategy 3 – Provide SME with system access (Inspector views while SME accesses the DVT)	
• Presents records and data directly within the system of record • Low effort to prepare • Ability to quickly adapt to changing requests • Familiar process, similar to presenting printed reports and data • Demonstrates transparency and builds trust with the inspector	• SMEs must be well versed in both the DVT and the content to be presented • SMEs must be confident in knowing how to present the information, and when to request an additional SME
Strategy 4 – Provide PDFs outside of the system (inspector is provided pdf printouts only)	
• Highest level of control for the organization • Can ensure content is reviewed by the audit team before presentation	• Dependent on system functionality and configuration to ensure full data is included • Volume of documentation can be prohibitive/difficult to navigate • Presents records and data outside of the system of record • Does not allow inspector to observe the DVT in use • Inspectors may not accept this strategy and may require the ability to directly view a system of record

It is recommended that organizations pursue Strategy 3 for presentation of a DVT, and the content within the DVT, whenever possible as this strategy strikes the best balance of control, transparency, effort, and flexibility.

7.3 Success Stories and Unexpected Challenges of Digital Validation and Inspection Implementation

7.3.1 Success Stories

7.3.1.1 Enhanced Compliance

DVTs facilitate compliance with regulatory requirements such as FDA 21 CFR Part 11 [1] and EU Annex 11 [7]. These tools help regulated companies streamline their validation process improvements and efficiencies while ensuring security, authenticity, and traceability of electronic records. Key features such as electronic signatures, audit trails, and data encryption ensure that organizations are prepared for regulatory inspections and can effectively demonstrate compliance.

To declare the implementation of a DVT a success, life science companies should observe enhanced compliance and noticeable increases (compared to traditional paper-based validation processes) as outlined below:

- Efficiency
- Real cost savings
- Improved data integrity
- Improved collaboration
- Faster access to information
- Real-time visibility of key metrics

7.3.1.2 Increased Efficiency

DVTs streamline and automate validation tasks, reducing the reliance on paper and manual data entry. This leads to faster turnaround times, improved productivity, and less administrative burden on validation teams.

Previously, validation efforts required considerable time and effort to compile aggregate data using templates. These efforts created significant barriers to change due to the high energy required to make the change. With a DVT, data collection and reporting are streamlined, allowing validation to occur closer to the person performing the action. This minimizes the need for large, time consuming reviews. For example, it is much easier for someone to review a short paragraph repeatedly over the course of a week than to find time to read a longer, more detailed document all at once.

Many DVTs feature independent requirement lifecycles, which allow for the seamless connection of many of the data elements such as testing, change requests, releases, related requirements, bugs, and other issues. This connection automatically creates a robust traceability matrix by default in real-time as the deliverables are being created, which not only reduces the effort but also simplifies investigation. For example, with the click of a button, users can easily view associated test cases and bugs linked to any requirement as opposed to going through a traceability matrix and then combing through test plans, a stack of executed tests, and defect trackers. The efficiency extends beyond creating the software, it also simplifies the outcome review and audit presentation. The entire system of operating becomes more efficient since data is captured as it happens, at the same moment that information was created.

7.3.1.3 Cost Savings

By reducing reliance on paper-based documentation, organizations can realize significant cost savings associated with paper, printing, storage, and document management. Additionally, streamlined validation processes can result in lower labor costs and improved resource utilization.

7.3.1.4 Improved Data Integrity

DVTs enhance data integrity by reducing the risk of errors, omissions, and inconsistencies commonly associated with manual data entry and paper-based documentation. Electronic records are easier to manage, track, and audit, ensuring the accuracy, completeness, and reliability of validation data.

Furthermore, as processes improve when using DVTs there is an increased capability for rule-based logic that can trigger actions or alerts if discrepancies occur. For example, if a user tries to mark a requirement state as complete without associated testing appearing to have been done, an alert will be created, notifying Quality for review and resolution, often finding that testing was completed, but the requirement was not associated with the test case when it was written. This can be explained in the comments by the user and QA, the testing can easily be linked and thus everything has been packaged nicely for any future investigations with very little effort. If testing is incomplete, the system could automatically send the requirement to a pending state with comments to complete testing and link it to an issue of requirements to be double checked prior to release.

7.3.1.5 Improved Collaboration

Electronic validation tools enable seamless collaboration among validation team members, stakeholders, and external partners. Users can access and collaborate on validation documents from anywhere, facilitating communication, feedback, and decision-making across geographically dispersed teams.

7.3.1.6 Faster Access to Information

Electronic validation tools allow users to quickly search, retrieve, and access validation documentation, reducing the time and effort required to locate and review paper-based records. This enables faster decision-making, troubleshooting, and problem resolution during validation activities.

7.3.1.7 Real-Time Visibility of Key Metrics

DVT is not just limited to final state or comparison if redlined previous versions. Software development tools and DVTs can maintain robust transaction logs and technical system audit trails. These logs capture actions in a readable way, providing much more insight into what happened, when it happened, and the reasoning because it captured the information from the transition between states as that information was created (refer to Section 7.3.1.2). This creates greater adherence to controls because many companies have already instituted single-sign-on, meaning the access to these tools is connected to a user's core credentials and thus brings a high-degree of assurance that the right individuals performed specific actions.

Additionally, DVTs can bring in dashboard capabilities that allow the team to package information to create an improved experience when reviewing or inspecting information.

7.3.2 Unexpected Challenges – Stories from the Front Line

This Guide focuses on the many benefits of implementing DVTS; however, from discussions with seasoned professionals who have implemented numerous systems in the life sciences industry, there are several unexpected challenges that organizations often face when transitioning from paper-based to digital systems, such as:

Legacy System Integration

Since DVTs often need to integrate with existing legacy systems, organizations may encounter compatibility issues, data migration complexities, and interoperability challenges. This can prolong implementation timelines and increase project costs.

Additionally, if a regulated company decides to divest some of its operations, it is important to ensure that the data integrity of validation records remains intact and accessible, especially if the DVTs validation tools are integrated with other systems. Interfaces may stop functioning and data integrity might not be prioritized during the divestment process. Therefore, a proper change control system and risk management are necessary to effectively mitigate these potential risks.

User Training and Adoption

Effective user training and adoption are critical for the successful implementation of a DVT. Ensuring that users are proficient in using the new system demands comprehensive training programs, ongoing support, understanding how to view data in the system and user-friendly interfaces to facilitate adoption and mitigate resistance. Without sufficient training and a deliberate roll out plan, the implementation is likely to fail, regardless of the new system capabilities.

Changes in Workflow

The transition to a DVT often necessitates a fundamental shift in established workflows and procedures. This transformation goes beyond a technical migration to a complete reimaging of how validation activities are conducted and managed.

One of the key challenges that organizations face is aligning existing validation workflows with the capabilities and functionalities of the new DVT. Conducting a thorough assessment of current processes to identify inefficiencies, redundancies, and opportunities for improvement is key to optimizing processes, streamlining validation processes, optimizing resource allocation, and eliminating bottlenecks that may impede productivity.

The shift to a DVT opens up new possibilities for collaboration and coordination among cross-functional teams. With a centralized digital platform, stakeholders can seamlessly collaborate on validation activities in real-time, fostering greater communication, transparency, and accountability across the organization. This collaborative approach fosters a culture of teamwork and innovation, driving continual improvement and driving organizational success.

Updates to Existing Quality Management System

The transition to a DVT requires reevaluating documentation practices and procedures. Electronic management of validation documents calls for robust document management procedures to ensure data integrity and compliance. By standardizing documentation practices and ensuring data integrity, organizations can ensure the compliance, accuracy, and reliability of validation records.

7.3.3 Success During Audits

While DVTs provide significant advantages to the validation process compared to the traditional paper-based processes, a critical element to address is how suitable it is for use during an audit. Therefore, consideration will need to be given to the following:

- The system should present the validation deliverables in a clear, understandable layout and format. This ensures that auditors can easily navigate through the documentation, understanding the validation process in order to review the necessary information.
- The system should allow for easy printing of the validation deliverables without requiring auditors to interact with the system itself. This ensures that auditors have the option to review printed documentation during audits, providing flexibility in how they access and review validation records.
- The system must clearly demonstrate who approved each deliverable and when. This includes capturing electronic signatures or approvals, along with timestamps, to provide a clear audit trail of the approval process.
- The system should clearly demonstrate the change control process. This includes documenting any changes made to validation deliverables, the nature of the change, the reason for the change, and the individuals involved in approving and implementing the change.

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The following outlines what life science companies can expect when using a DVT during audits:

Compliance with Regulatory Requirements

Auditors will evaluate whether the DVT complies with relevant regulatory standards including FDA 21 CFR Part 11 [1], EU Annex 11 [7], and other applicable guidelines. They will review documentation, electronic signatures, audit trails, and system validation records to ensure full compliance.

Validation Documentation for the DVT

The validation documentation will be reviewed to verify that the validation efforts are appropriate, current, and accurately reflect the system's intended use and functionality.

As outlined in this Guide and aligning with GAMP validation principles, critical thinking should be applied during the validation planning process. The focus should be on ensuring that compliance checks are functioning as intended.

In connection with the validation process, expect the change control process to be evaluated to ensure that any changes, updates, or modifications to the system are properly documented, assessed for impact on validation, and implemented in a controlled manner to prevent unintended consequences.

Vendor Management

As many of these systems are provided by third-party vendors, auditors will review the company's vendor management practices. They will evaluate the vendor's qualifications, contracts, and SLAs to ensure that the vendor meets regulatory requirements and offers adequate support for the system. This is especially important if vendor testing is leveraged to reduce the validation burden during DVT implementation.

Training and Competency

Auditors will assess whether employees using the DVT are adequately trained and competent in its use by reviewing training records, competency assessments, and procedures to ensure that users understand how to effectively operate the system and remain compliant with regulatory requirements.

Contingency Planning

Auditors will evaluate contingency plans for managing system failures, data breaches, or other disruptions. They will determine whether the company has procedures in place to mitigate risk, restore system functionality, and preserve data integrity in the event of an incident.

Application of Critical Thinking

Regulated companies will need to demonstrate that critical thinking has been applied throughout the validation of the software. For example, there is an opportunity for users of a DVT to assess risks on a per requirement basis and potentially overlook broader system risks. Auditors will be interested in understanding this context and how critical thinking has been applied to the overall software validation process.

Presentation of Validation Deliverables

Careful consideration will need to be given to how the validation deliverables are presented within the DVT. Auditors typically expect to see complete documents, such as the entire user requirements specification or the risk analysis document. This can be challenging as more life science companies move towards an Agile lifecycle with a focus on presenting sets of data/information to support validation. The business owner should be prepared to present deliverables in a way that demonstrates reduced risk and greater visibility.

Accessibility and Organization

Auditors expect validation deliverables to be easily accessible and well-organized within the DVT. The system should provide clear navigation and search functionalities to help auditors locate specific documents, such as validation protocols, test scripts, and validation reports.

Document Control

Auditors will scrutinize the document control practices within the DVT to ensure that all validation documents are properly controlled, versioned, and accessible only to authorized personnel. They will verify that the system maintains an audit trail for document changes and has mechanisms in place to prevent unauthorized alterations.

Document Formatting and Completeness

Validation documents should follow standard formats and templates to ensure consistency and completeness. Auditors will review the content to confirm that they contain all required information, including objectives, scope, methodologies, results, conclusions, and signatures.

Electronic Signatures and Approvals

Auditors will assess the management of electronic signatures and approvals within the DVT. Data review may include an evaluation of timelines to determine if approvals/signatures were obtained before the relevant activities commenced.

Integration with Other Systems

If the DVT integrates with other enterprise systems, auditors will evaluate how data exchanges between systems are managed and controlled. They will assess whether data exchanges are well-documented, validated, and secure to prevent data loss, corruption, or unauthorized access.

By addressing these concerns and ensuring that effective presentation of deliverables within the DVT, companies can demonstrate their commitment to compliance, transparency, and regulatory integrity. This approach will help ensure that the DVT successfully supports an audit.

7.4 Future Trends and Developments

7.4.1 Emerging Technologies in Digital Tools

Emerging technologies in digital tools are revolutionizing how organizations manage, process, and utilize information. There are many benefits as well as associated risks. Here are some of the key technologies driving this transformation:

Cloud Computing

Provides scalable and on-demand access to computing resources and storage.

- Benefits: Enhances collaboration, remote access, simplifies data backup, and reduces reliance on physical storage

Artificial Intelligence (AI) and Machine Learning (ML)

AI and ML algorithms can automate data processing, analysis, and decision-making such as protocol generation from requirements and suggesting risk mitigations strategies.

- Benefits: Improves accuracy, consistency, reduces manual labor, and provides predictive analytics for proactive management

Big Data Analytics

Enables the processing of large volumes of data to extract actionable insights.

- **Benefits:** Enhances data-driven decision-making, identifies trends, and patterns, and improves forecasting

Electronic Signature (eSignature)

eSignatures offer a secure and legal way to sign documents electronically and continue to evolve with emerging technologies

- **Benefits:** Accelerates the signing process, reduces paper usage, and enhances security and compliance

Virtual and Augmented Reality (VR/AR)

Immersive technologies that create virtual environments or overlay digital information onto the real world.

- **Benefits:** Enhances training, simulations, and remote assistance without the need for physical documentation

Emerging technologies in digital tools come with several risks, including:

- **Cybersecurity Threats:** Increased vulnerability to hacking, data breaches, and malware.
- **Privacy Concerns:** Potential misuse of personal data and surveillance risks.
- **Bias and Ethical Issues:** AI and automation can reinforce biases and lead to unfair outcomes.
- **Regulatory and Compliance Risks:** Rapid tech advancements may outpace laws and regulations.
- **Dependence on Technology:** Over-reliance can lead to operational failures if systems crash.
- **Misinformation and Deepfakes:** AI-generated content can spread false or misleading information.

By adopting these emerging technologies, organizations can significantly reduce their reliance on paper, improve operational efficiency, and enhance their ability to manage and leverage data. When implementing these technologies, consider the following:

- **Security:** Implement robust data security and privacy measures to protect sensitive information
- **Compliance:** Ensure adherence to industry regulations and standards for digital records and electronic transactions
- **Integration:** Ensure new technologies can integrate seamlessly with existing systems and workflows
- **User Training:** Provide comprehensive training to users to maximize the benefits and minimize resistance to new technologies

8 Appendix 1 – References

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9 Appendix 2 – Glossary

9.1 Acronyms and Abbreviations

AI	Artificial Intelligence
ALCOA++	Attributable, Legible, Contemporaneous, Original, and Accurate, Complete, Consistent, Enduring, Available, and Traceable
BCP	Business Continuity Plan
C&Q	Commissioning and Qualification
CDE	Critical Design Element
CDMO	Contract Development and Manufacturing Organization
CMMS	Computerized Maintenance Management System
COP	Community of Practice
CPP	Critical Process Parameter
CPV	Continuous Process Verification
CQV	Commissioning, Qualification, and Validation
CSA	Computer Software Assurance
DCS	Distributed Control System
DMAIC	Define, Measure, Analyze, Improve, and Control
DMS	Document Management System
DMT	Deviation Management Tool
DV	Digital Validation
DVT	Digital Validation Tool
EIMS	Engineering Information Management System
ELB	Electronic Logbook
FAQ	Frequently Asked Questions
FAT	Factory Acceptance Testing
FDA	Food and Drug Administration (US)
GDocP	Good Documentation Practice
GDP	Good Distribution Practice
GEP	Good Engineering Practice
GMP	Good Manufacturing Practice
GxP	Good “x” Practice
IaaS	Infrastructure as a Service
ICH	International Council for Harmonisation
IP	Intellectual Property
IQ	Installation Qualification

IT	Information Technology
KPI	Key Performance Indicator
LIMS	Laboratory Information Management System
LLM	Large Language Model
LMS	Learning Management System
MDIC	Medical Device Innovation Consortium
ML	Machine Learning
MRP	Materials Requirements Planning
NER	Named Entity Recognition
NOR	Normal Operating Range
OCR	Optical Character Recognition
OQ	Operational Qualification
P&ID	Piping and Instrumentation Diagram
PaaS	Platform as a Service
PUR	Process User Requirement
PV	Process Validation
QA	Quality Assurance
QC	Quality Control
QMS	Quality Management System
QRM	Quality Risk Management
ROI	Return on Investment
RS	Requirement Specification
SaaS	Software as a Service
SaMD	Software as a Medical Device
SLA	Service Level Agreement
SME	Subject Matter Expert
SOP	Standard Operating Procedures
SST	System Suitability Testing
UAT	User Acceptance Testing
UNS	Unified Namespace
URS	User Requirement Specification
VR	Virtual Reality

9.2 Definitions

Digital Ecosystem

An integrated connection of the different software's and automation systems enabling data flow.

Digital Validation Tool (DVT)

A range of software applications (one or more integrated) designed for managing and overseeing digital assets essential for the qualification, verification, validation, and assurance of compliance in various life sciences domains.

Paper on Glass

A process of implementing current paper-based processes into digital form without optimizing to a digital approach.

Process Owner

This is the owner of the business process or processes being managed. The process owner is ultimately responsible for ensuring that the computerized system and its operation is in compliance and fit for intended use, in accordance with the applicable company Standard Operating Procedures (SOPs). The process owner may also be the system owner. The process owner may be the *de facto* owner of the data residing on the system (data owner) and therefore, ultimately responsible for the integrity of the data. Process owners are typically the head of the functional unit using the system.



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